ETH zürich

Mobile Computing

Foundations of Wireless Communications

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Course Outline

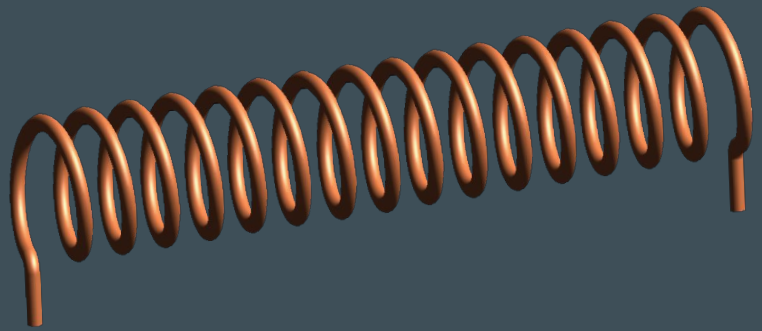
Introduction

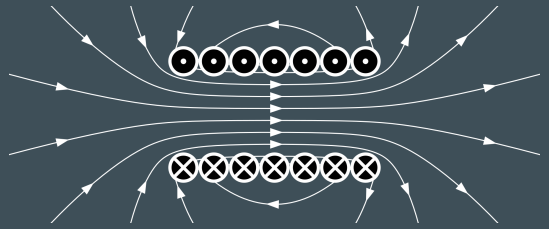
1. Data and Signals
2. Data Communication
3. Mobile Systems
4. Mobile Algorithms
5. Mobile AI
6. Cloud Computing

Tournament

Creating an Electromagnetic Wave

- Alternating electrical current, flowing through an inducting coil (I·R) induces a magnetic field B.
- This alternating field B induces an electric field E, which induces a magnetic field, which in return induces another electric field, and so on ...





Integral forms of Maxwell's Equations

1. Gauss's Law (Electric Field)

$$\oint_{\partial V} \mathbf{E} \cdot d\mathbf{A} = \frac{Q_{\text{enc}}}{\varepsilon_0}$$

   Electric flux through a closed surface is proportional to charge enclosed within that surface

2. Gauss's Law for Magnetism

$$\oint_{\partial V} \mathbf{B} \cdot d\mathbf{A} = 0$$

   No monopoles; magnetic flux through closed surface is zero

3. Faraday's Law of Induction

$$\oint_{\partial S} \mathbf{E} \cdot d\mathbf{l} = -\frac{d}{dt} \int_S \mathbf{B} \cdot d\mathbf{A}$$

   A changing magnetic field induces electric field (electromagnetic induction)

4. Ampère's Law

$$\oint_{\partial S} \mathbf{B} \cdot d\mathbf{l} = \mu_0 \left(I_{\text{enc}} + \varepsilon_0 \frac{d}{dt} \int_S \mathbf{E} \cdot d\mathbf{A} \right)$$

   Magnetic fields can be generated by electric currents and / or changing electric field

Time-varying Electric and Magnetic Fields

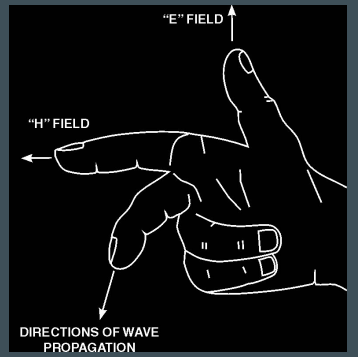
- When a charged particle moves or accelerates, it disturbs the electric and magnetic fields around it
  - These disturbances do not stay localized
  - Instead, they travel outward from the source in the form of an electromagnetic wave
- Fields regenerate each other as they move through space
  - A changing electric field creates a magnetic field
  - A changing magnetic field creates an electric field

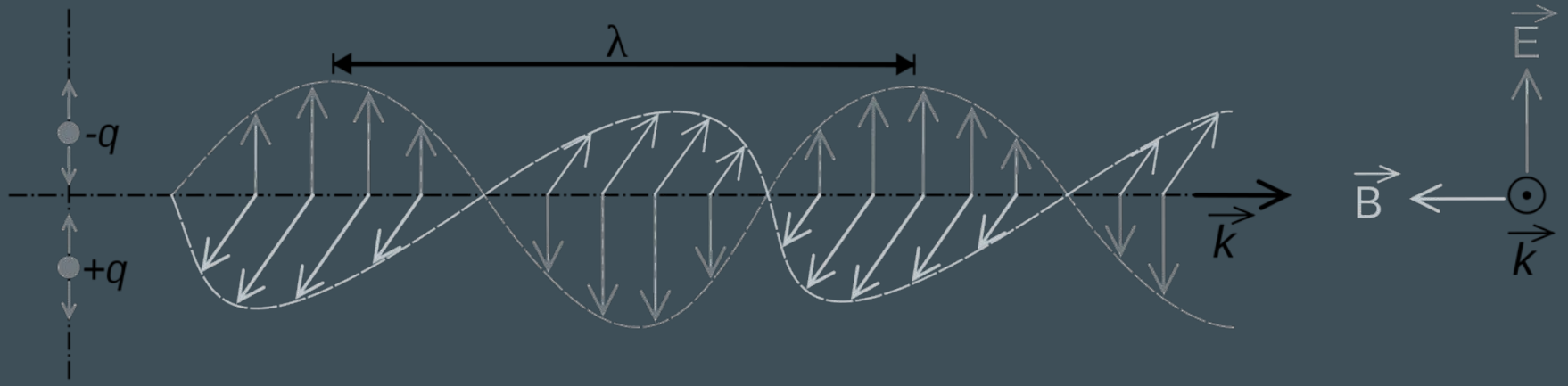
Time-varying Electric and Magnetic Fields

- A changing electric field creates a magnetic field
- A changing magnetic field creates an electric field
- Coupling between electric and magnetic fields leads to propagating electromagnetic wave

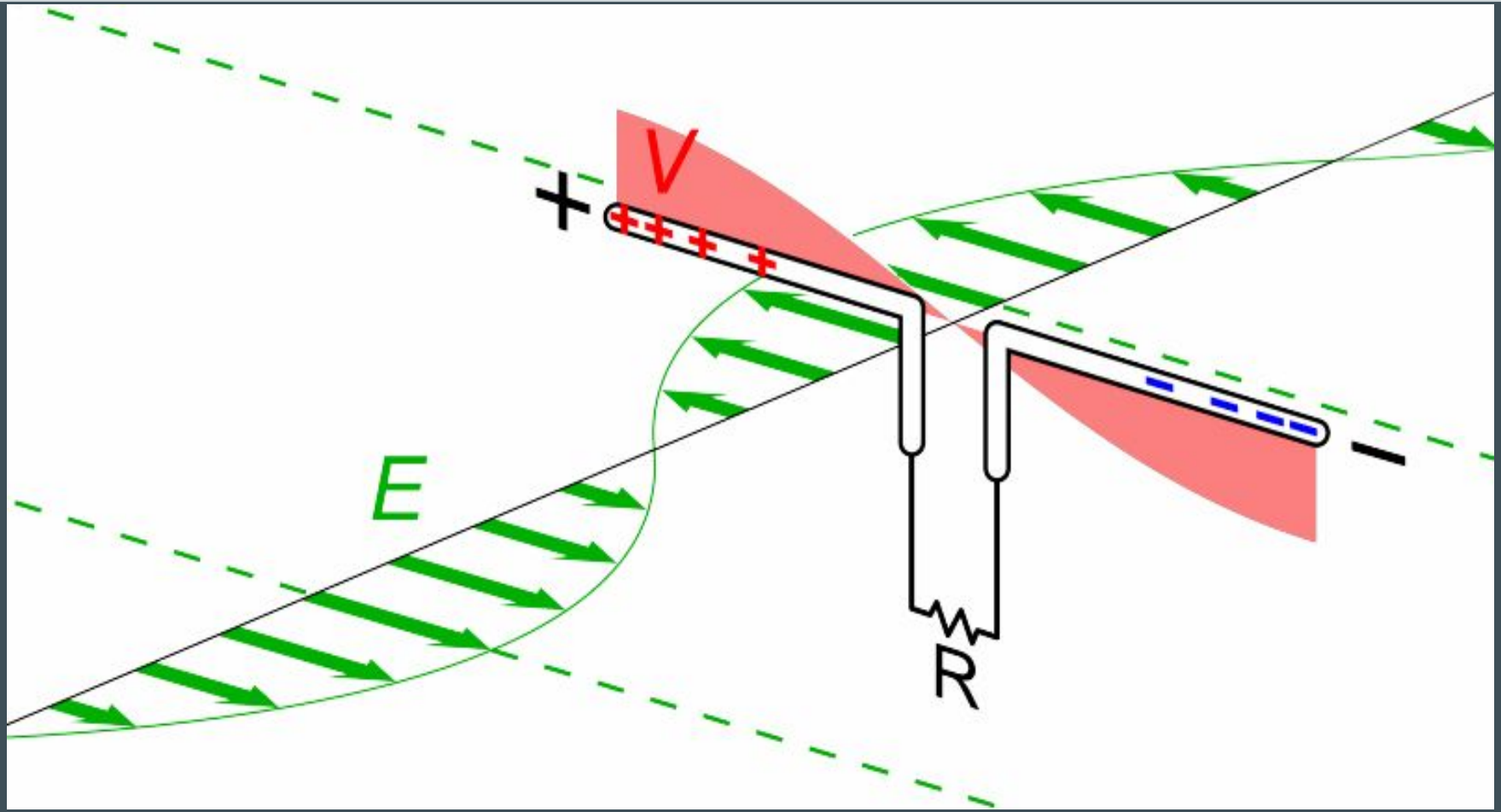
Radiation and Wave Propagation

- At high frequencies, electromagnetic waves propagate through medium (air)
  - Only far away from antenna
  - The graph below is a simplification

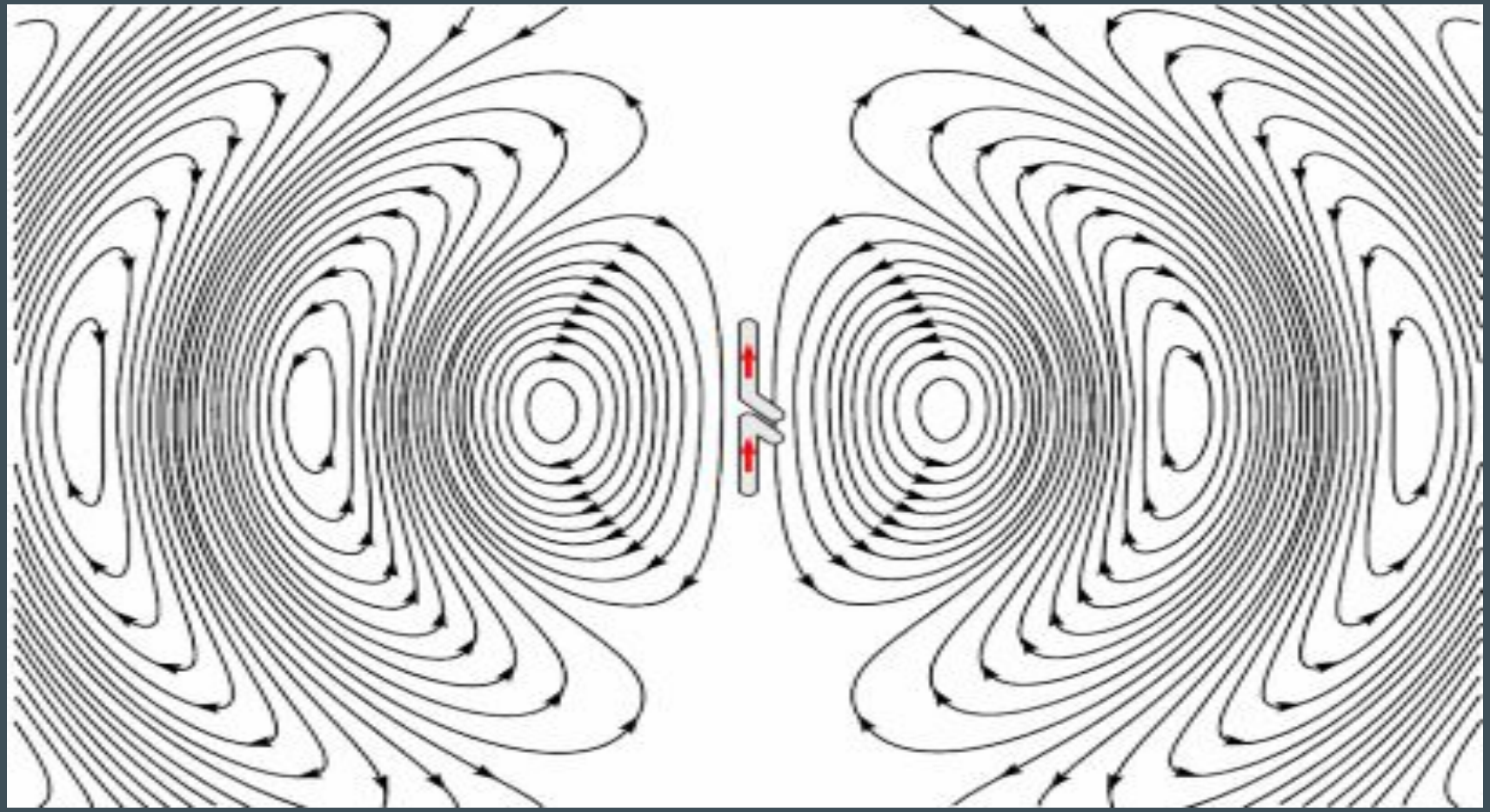




Dipole Antenna [Source]



Dipole Antenna [Source]

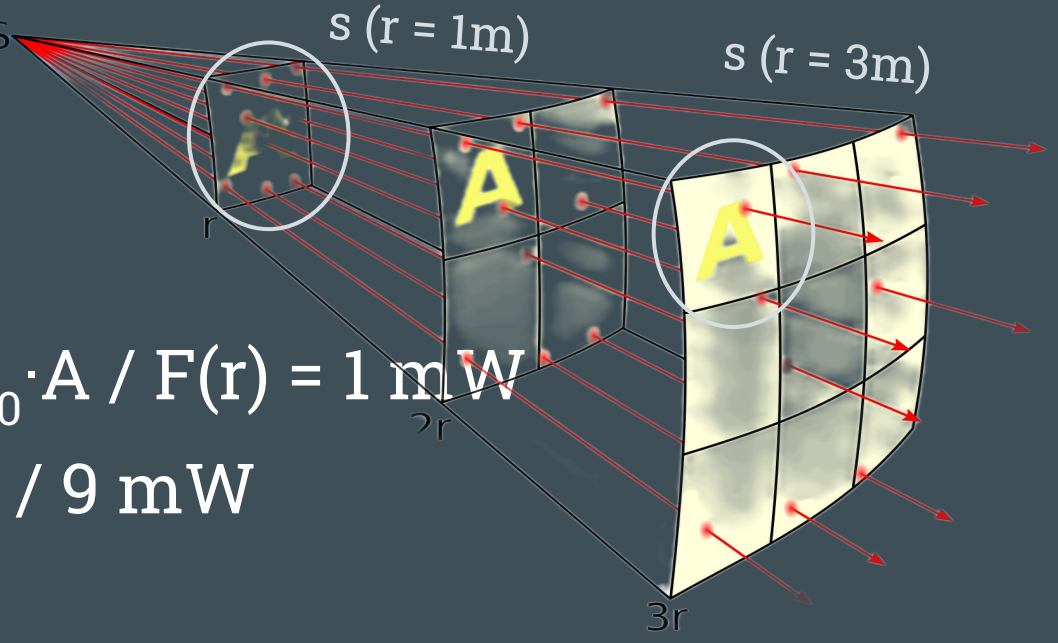


Loss of Signal Strength over Distance

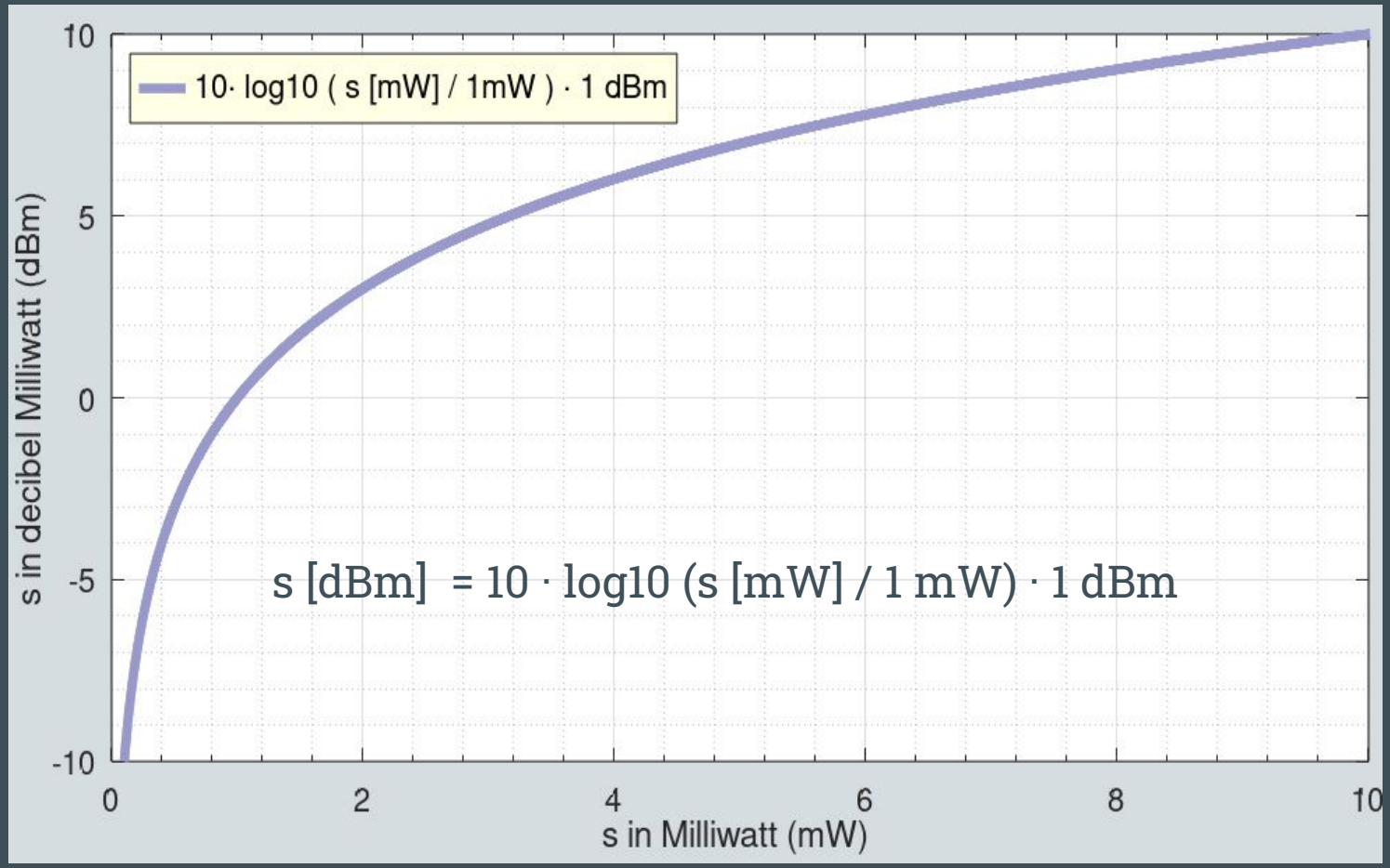
- Inverse-square law of signal strength: $P_{RX}(r) \sim 1 / r^2$
- Surface area of expanding sphere $F(r) = 4 \pi r^2$
- Antenna size A
- $s(r) = s_0 \cdot A / F(r)$
- Example

  $P_{RX}(r = 1m) = s_0 \cdot A / F(r) = 1$ mW

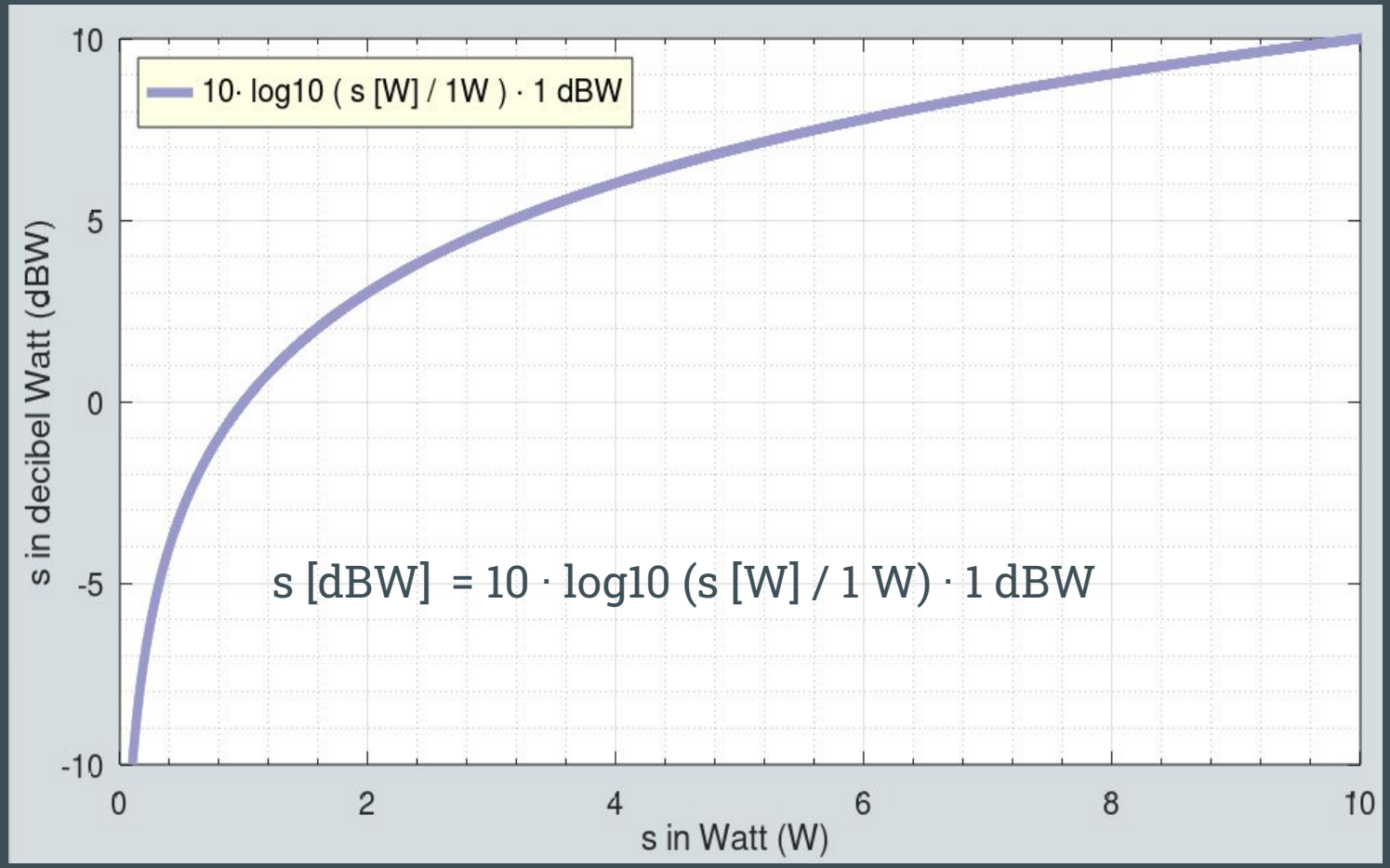
  $P_{RX}(r = 3m) = 1 / 9$ mW



Milliwatt (mW) and dBm



Watt (W) and dBW



Signal Power in dBm (or in dBW)

- $s\,[\text{dBm}] = 10 \cdot \log10\,(s\,[\text{mW}] / 1\,\text{mW}) \cdot 1\,\text{dBm}$
- $s\,[\text{dBW}] = 10 \cdot \log10\,(s\,[\text{W}] / 1\,\text{W}) \cdot 1\,\text{dBW}$
- $s\,[\text{dBm}] = 10 \cdot \log10\,(s[\text{Watt}] / 1\,\text{mW}) \cdot 1\,\text{dBm}$
  $= 10 \cdot \log10\,(s\,[\text{W}] / 0.001\,\text{W}) \cdot 1\,\text{dBm}$
  $= 10 \cdot (\log10\,(s\,[\text{W}] / 1\,\text{W})) + 30 \cdot 1\,\text{dBm}$
- $1\,\text{dBW} = 30\,\text{dBm}$
- Remark: $1\,\text{W} = 1000\,\text{mW}$

Signal Attenuation / Amplification

- Multiplying with a constant factor, for example:
  - $s_{received}[mW] = s_{transmitted}[mW] \cdot 0.001$
  - $s_{received}[dBm] = 10 \cdot (\log10 (s_{transmitted}[mW] / 1mW) - 30) \cdot 1\ dBm$
  - $s_{received}[dBm] = s_{transmitted}[dBm] - 30\ dBm$
- Other examples
  - Double a value: + 3 dB
  - Cut value in half: - 3 dB
  - Amplify a signal by factor 4: + 6 dB
  - Attenuate a signal by factor 10: - 10 dB
  - Attenuate a signal by factor 1000: - 30 dB

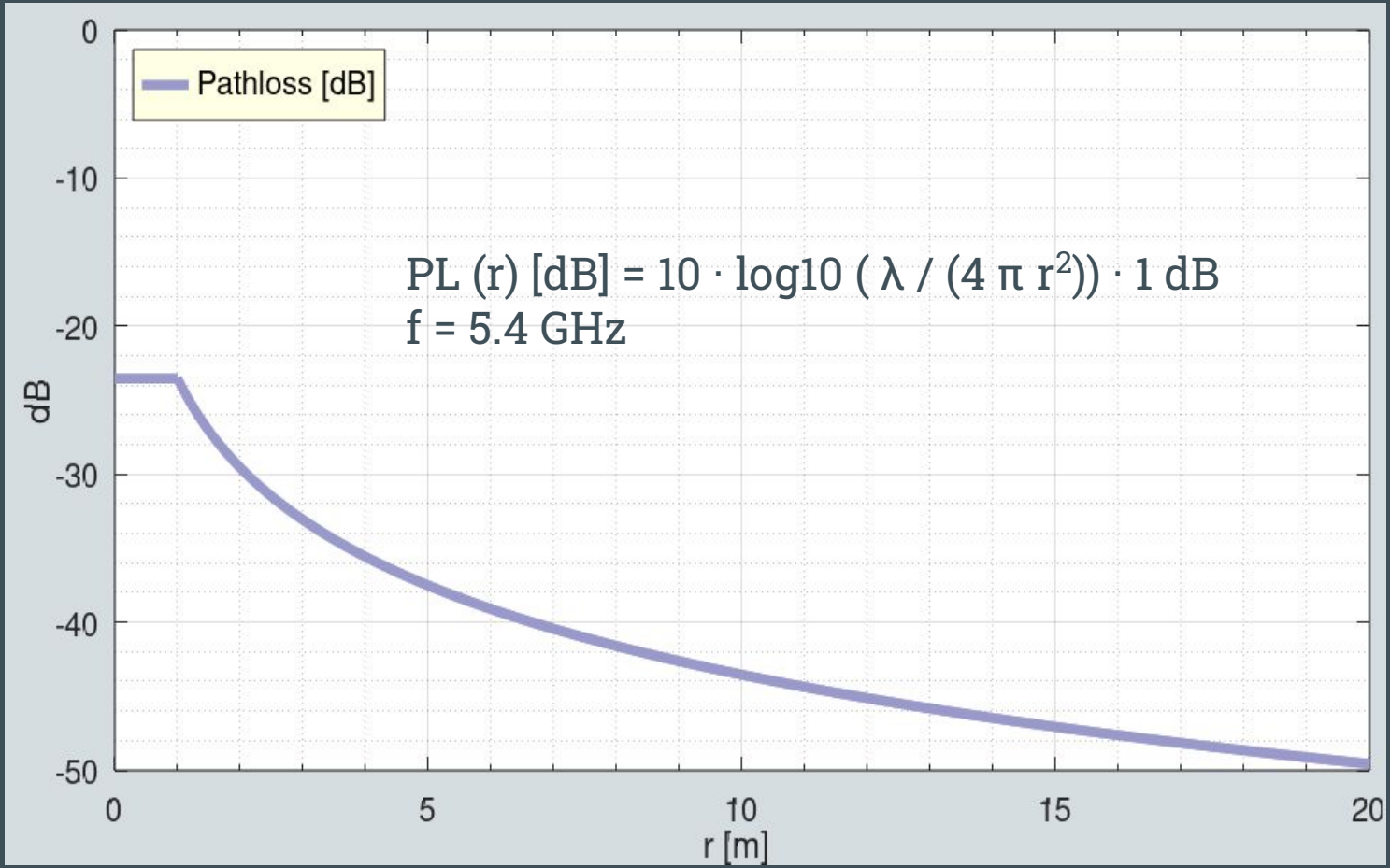
Pathloss [dB]

- $PL(r) = 10 \cdot \log10 (\lambda / (4 \pi r^2)) \cdot 1 dB$
- $PL(r) = 10 \cdot (\log10 (\lambda / (4 \pi)) - 2 \cdot \log10 (r / 1m)) \cdot 1 dB$
  - with wavelength $\lambda = c / f$ and distance r
- Power of the received signal s (r) in dBm:
  - $P_{RX}(r) = P (s_0) [dBm] + PL [dB]$
- Example: f = 2.4 GHz, and signal power $P (s_0) = 10$ mW
  - Path loss after one meter: PL (1 m) = - 20.023 dB
  - $P_{RX} (1 m) = P(s_0) + PL (1m) = 10 dBm - 20.023 dB = - 10.023 dBm$

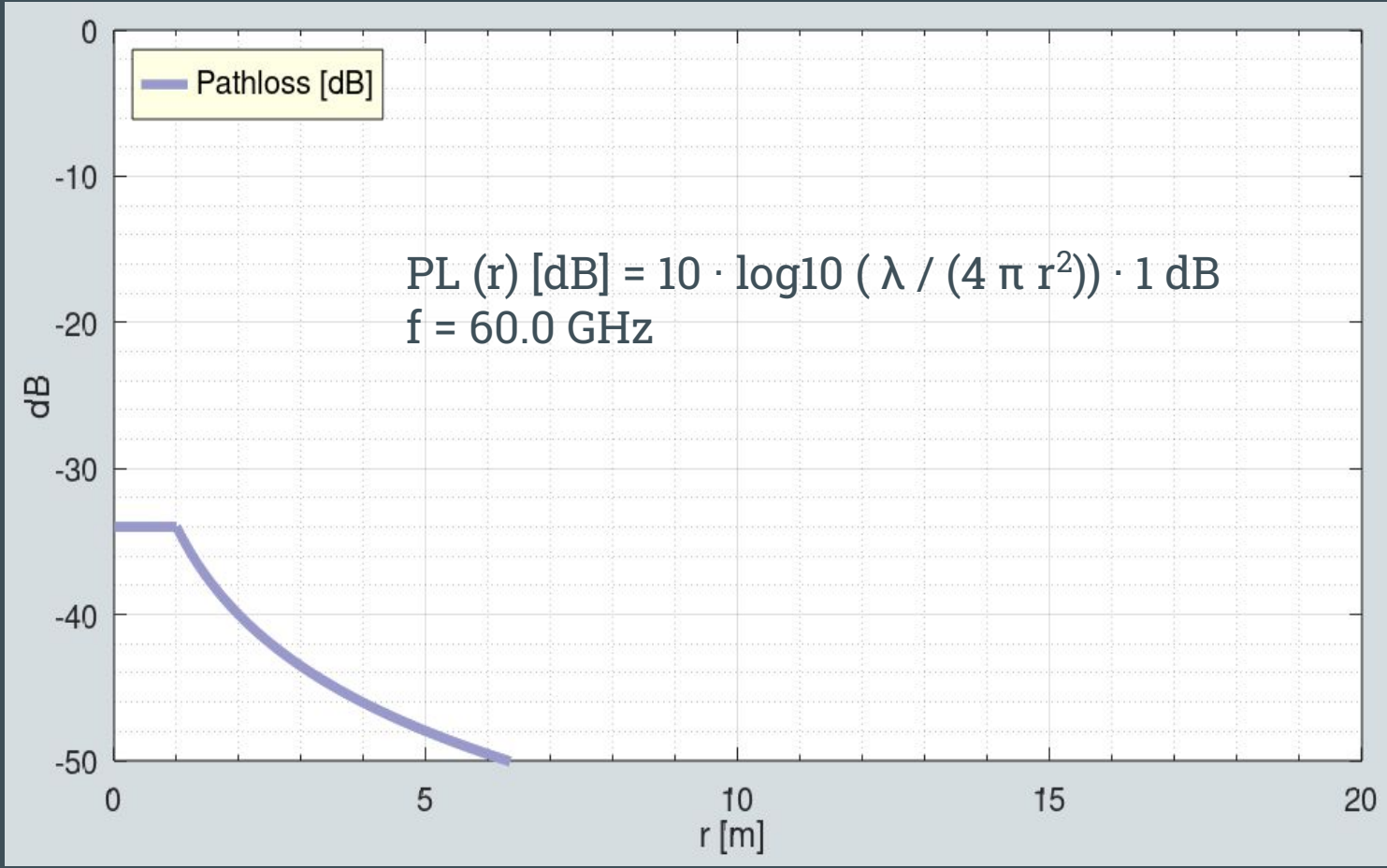
Path Loss at 2.4 GHz



Path Loss at 5.4 GHz



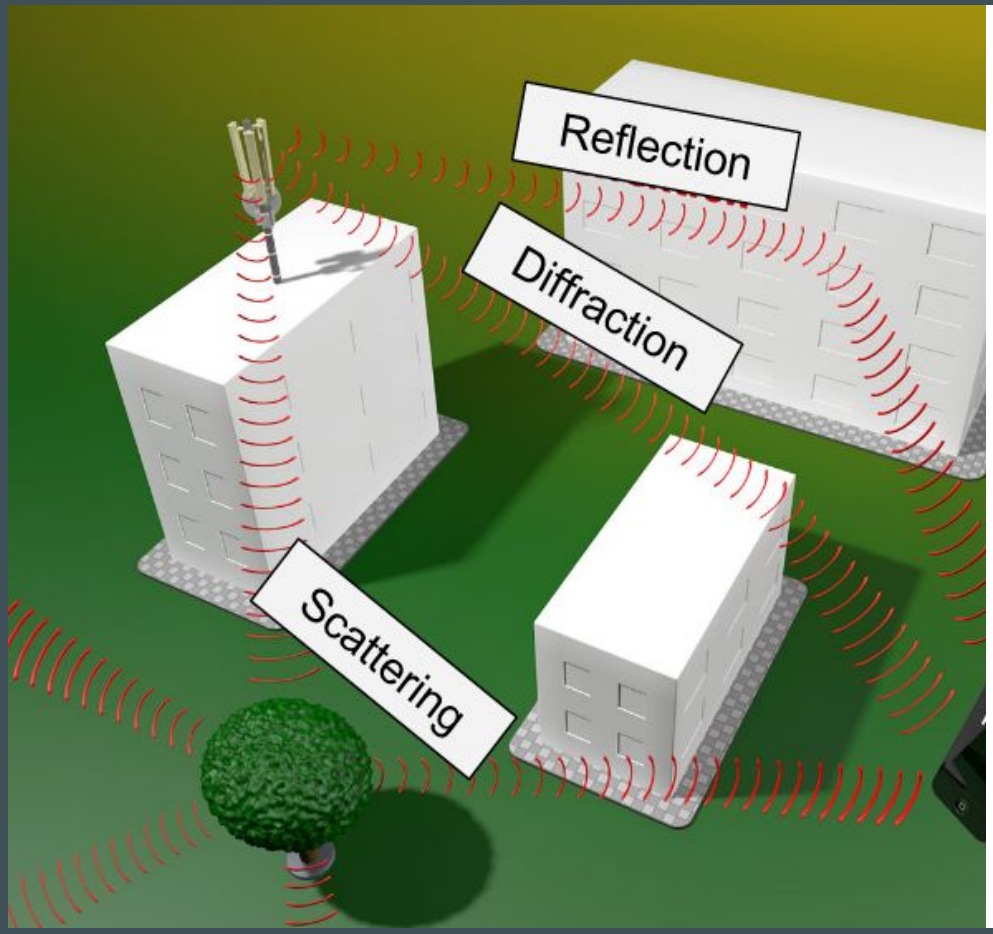
Path Loss at 60 GHz

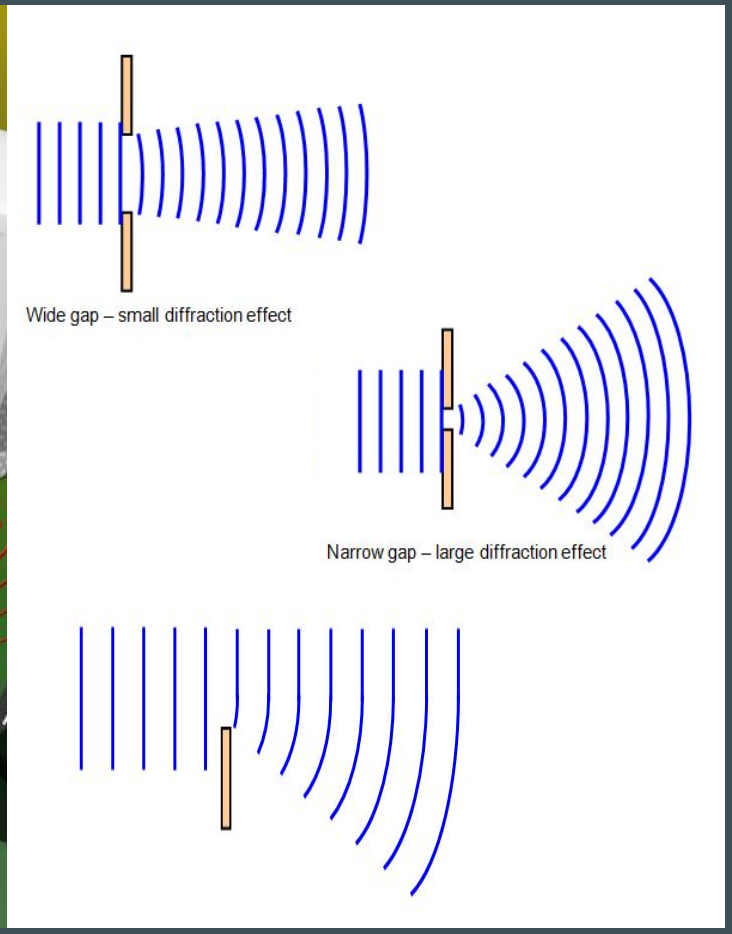


Path Loss Calculation in Jemula802++

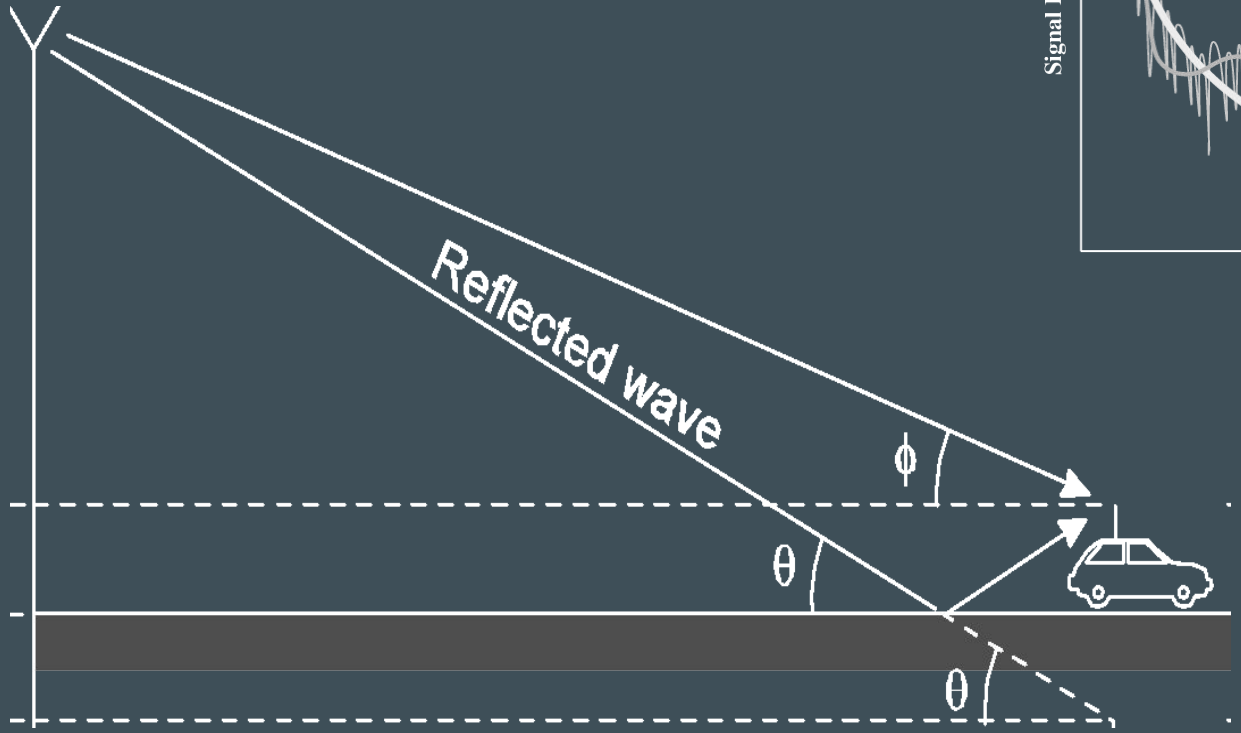
```
public double calcPathloss(...) {
  double lambda = 3E8 / (getDot11CenterFreq_MHz() * 1E6);
  if (distance > 1.0) {
anAttenuation_dB = getAttenuationCoeff() * 10 * Math.log10(lambda /
   (4 * Math.PI * distance));
  } else {
anAttenuation_dB = getAttenuationCoeff() * 10 *
Math.log10(lambda / (4 * Math.PI * 1 ));
  }
  return (anAttenuation_dB + srcDirectionalGain_dBi);
}
```

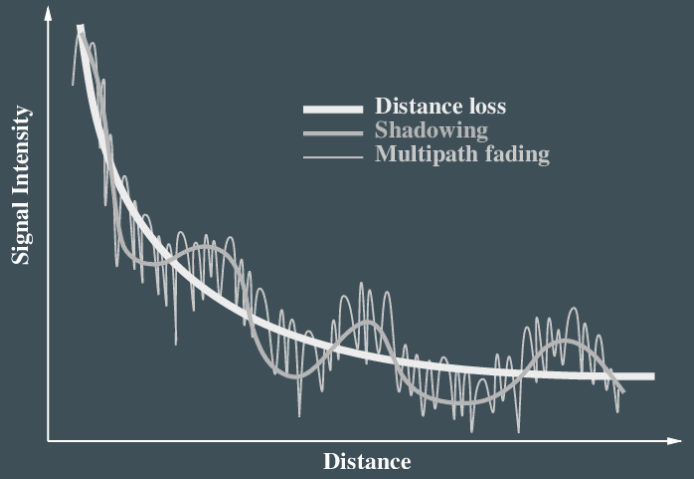
Diffraction, Reflection, Scattering





Shading and Multi-Path Propagation and Fading

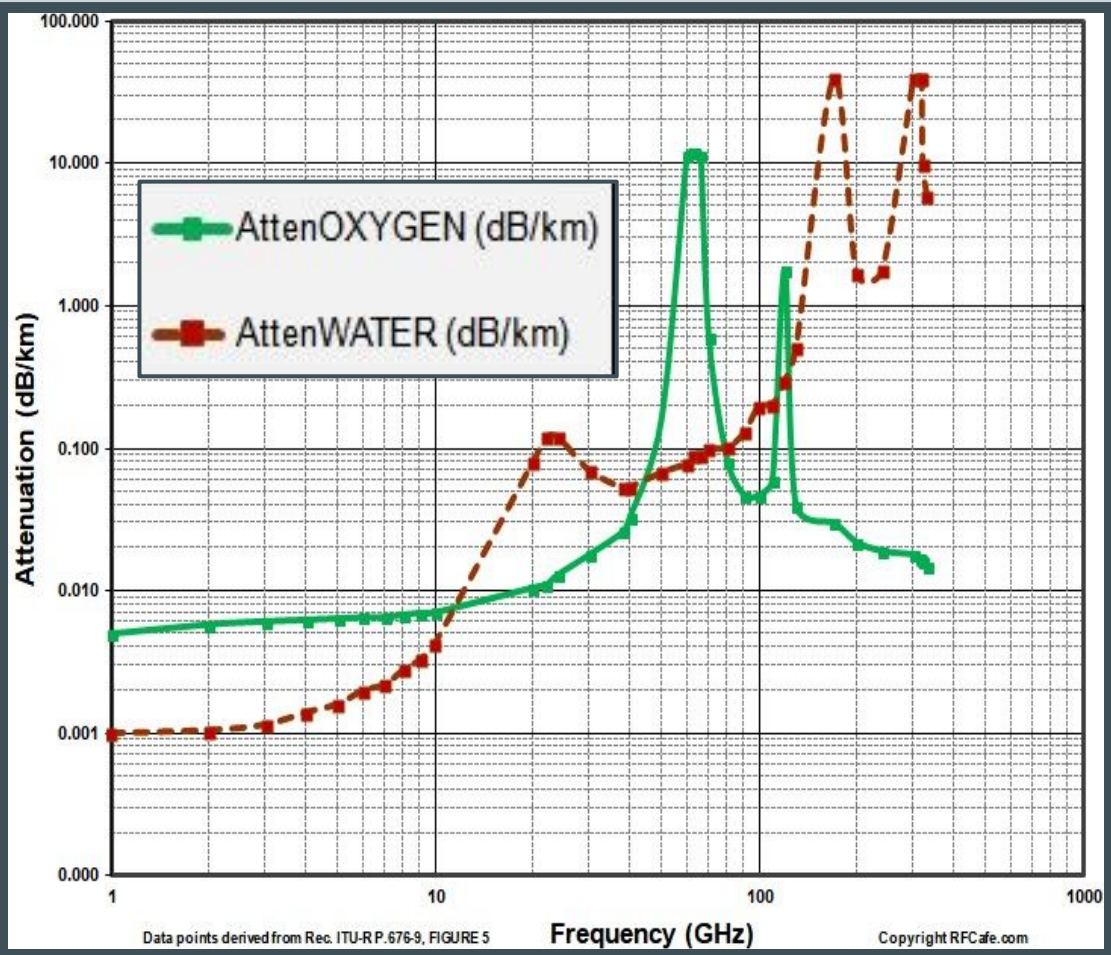




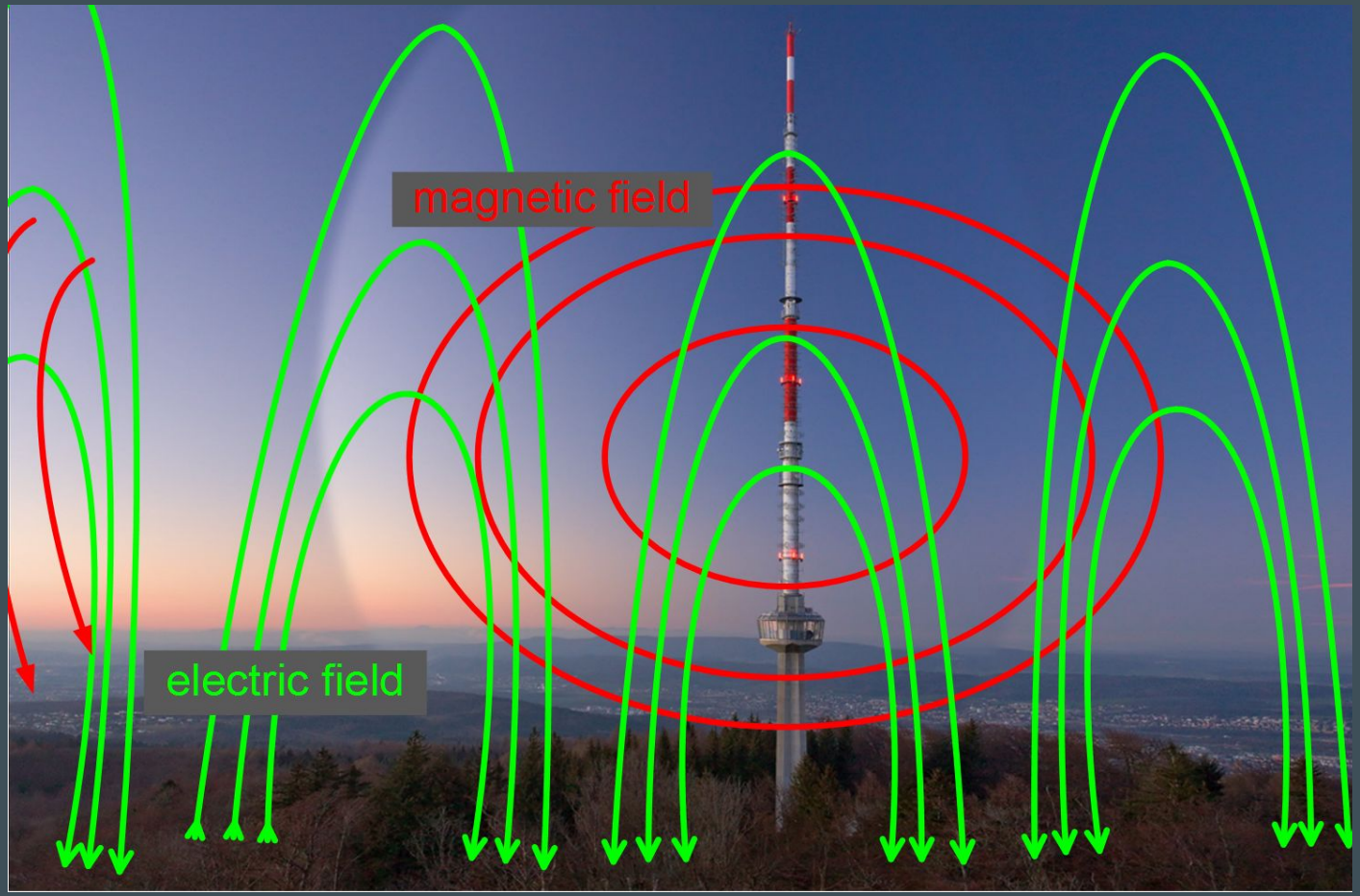
Our Atmosphere

- Nitrogen (N_2): ~78% - stable, inert, acts as buffer gas
- Oxygen (O_2): ~21% - for respiration and combustion
- Argon (Ar): ~0.93% - inert noble gas
- Carbon dioxide (CO_2): ~0.04% - for photosynthesis and climate regulation
- Water vapor (H_2O): 0-4% - highly variable; drives weather and climate
- Neon, Helium, Krypton, Hydrogen, Xenon - small traces

Attenuation through Atmosphere



Antenna Radiation Pattern



Antenna Feed



Parabolic Antennas



Radomes



Protected Antenna Systems



Parabolic Antenna Feed



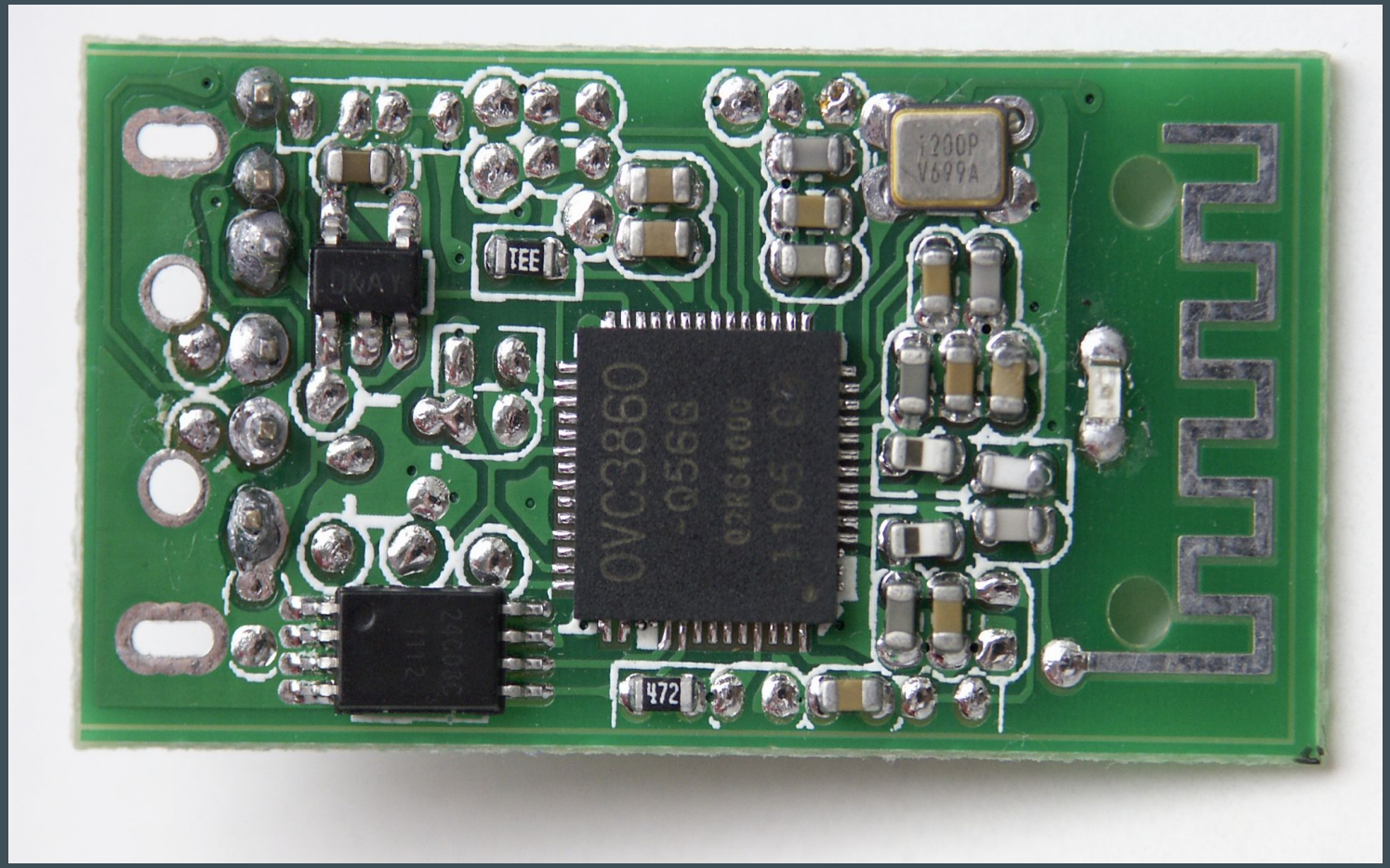
Satellite Dish



Cellular Base Station



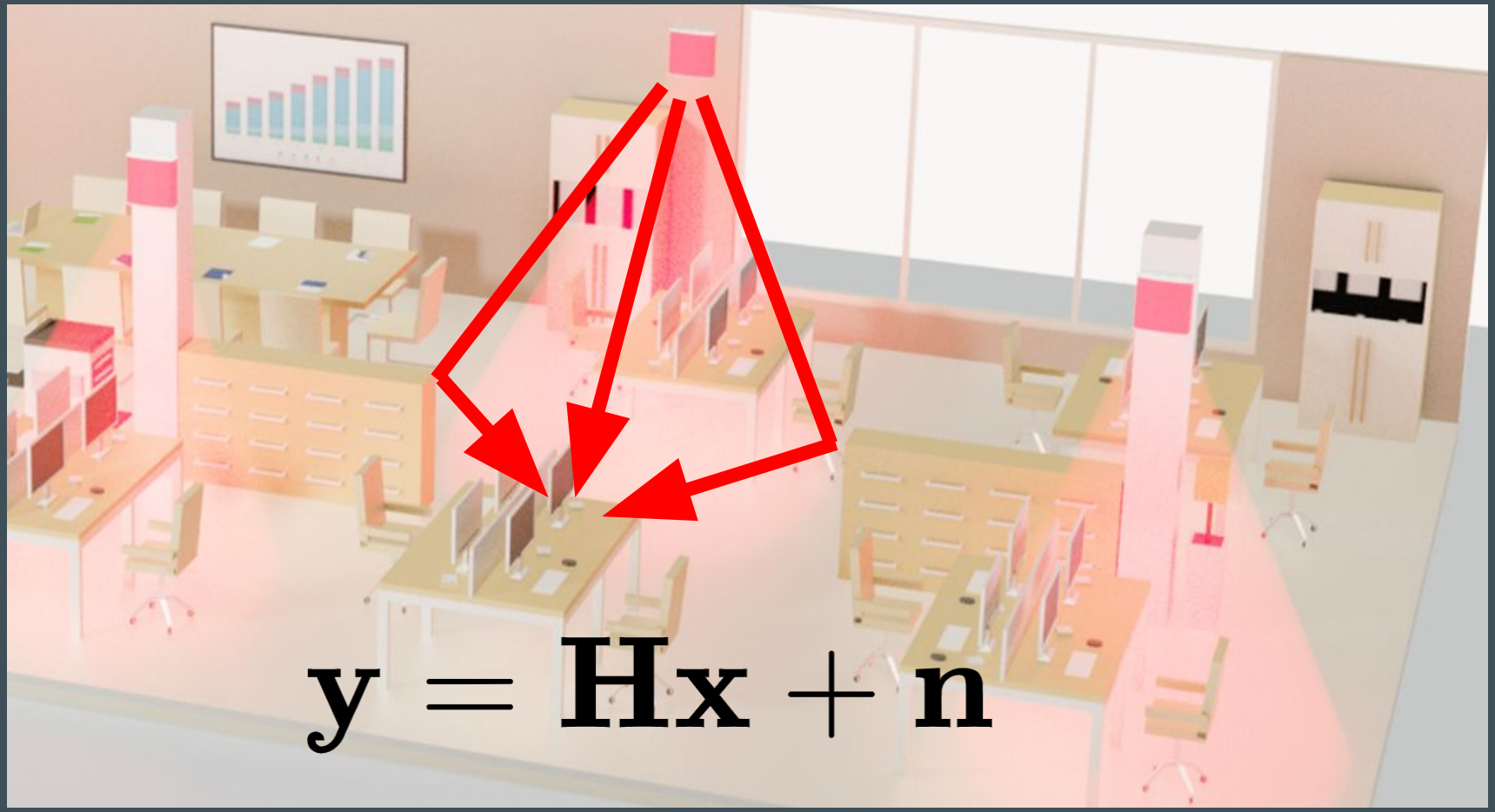
Antennas on a Circuit Board



Wi-Fi Antennas



MIMO (Multiple-Input Multiple Output)

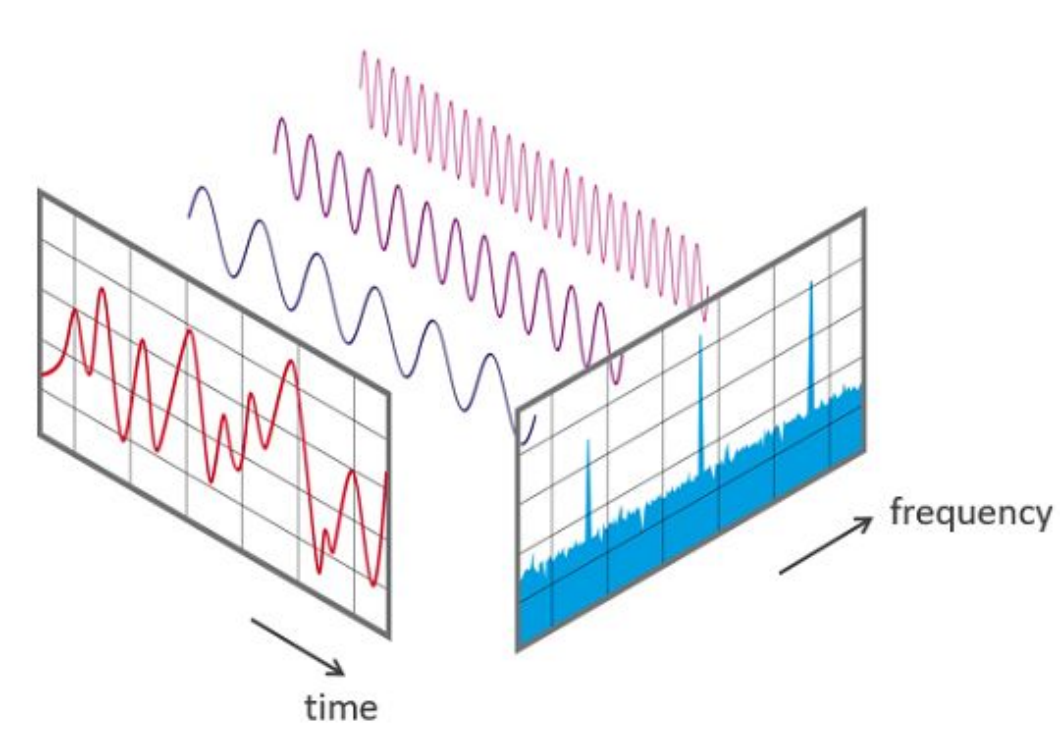


Independent MIMO Streams

- y is the vector of received signals
- H is the channel matrix
- x is the vector of transmitted signals
- n is the noise vector

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ y_3 \end{bmatrix} = \begin{bmatrix} h_{11} & h_{12} & h_{13} \\ h_{21} & h_{22} & h_{23} \\ h_{31} & h_{32} & h_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} + \begin{bmatrix} n_1 \\ n_2 \\ n_3 \end{bmatrix}$$

Signals in Time and Frequency Domain



Source: Fast Fourier Transformation FFT - Basics

Fourier Transform

- Time to Frequency

$$X(f) = \int_{-\infty}^{\infty} x(t) \cdot e^{-j2\pi ft}\, dt$$

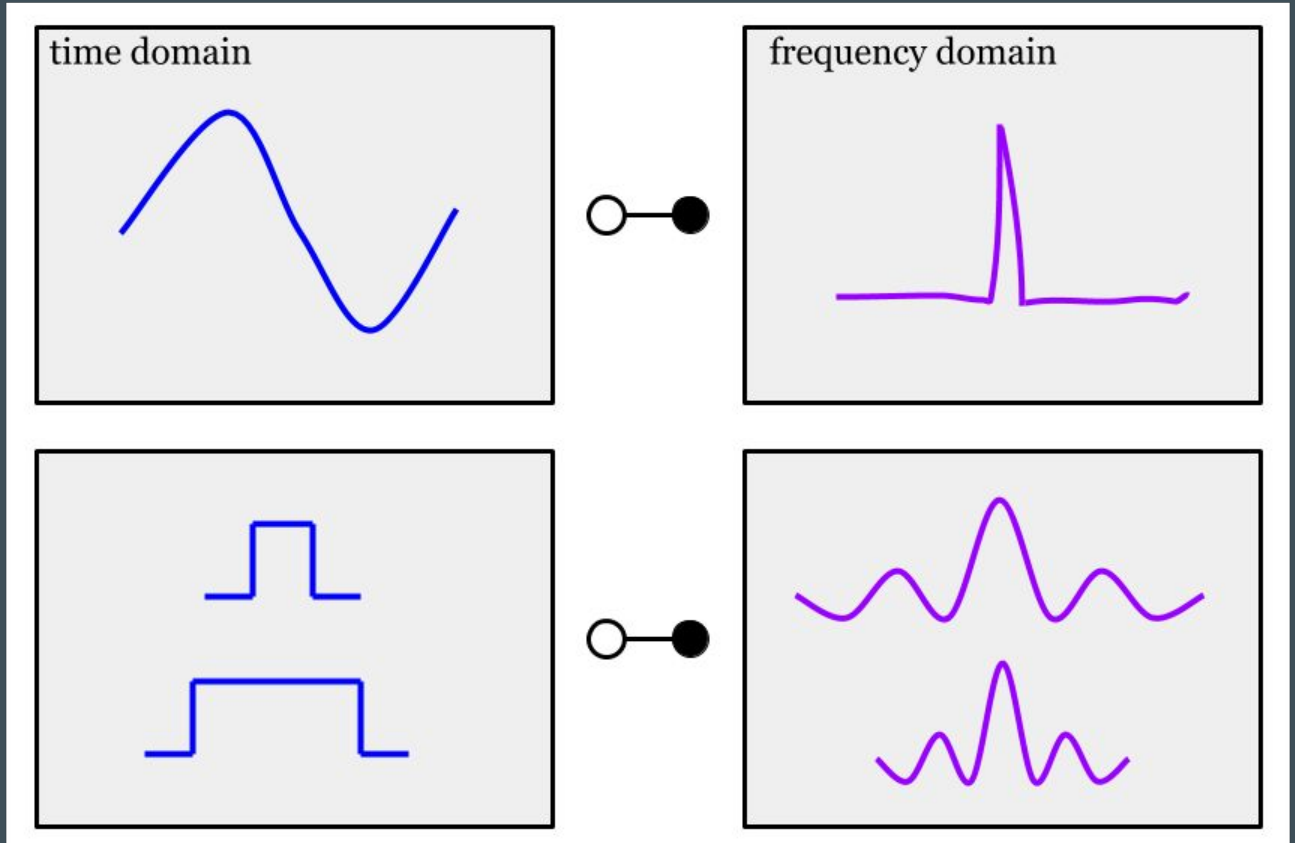
- Inverse Fourier Transform (Frequency to Time)

$$x(t) = \int_{-\infty}^{\infty} X(f) \cdot e^{j2\pi ft}\, df$$

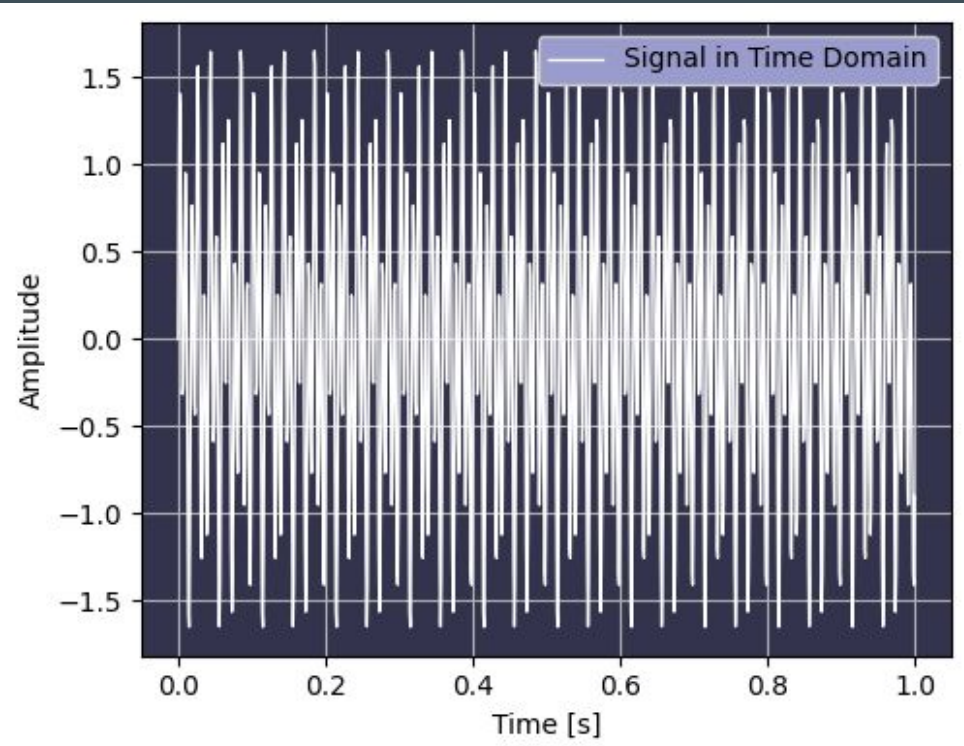
- Complex exponentials with Euler's Formula

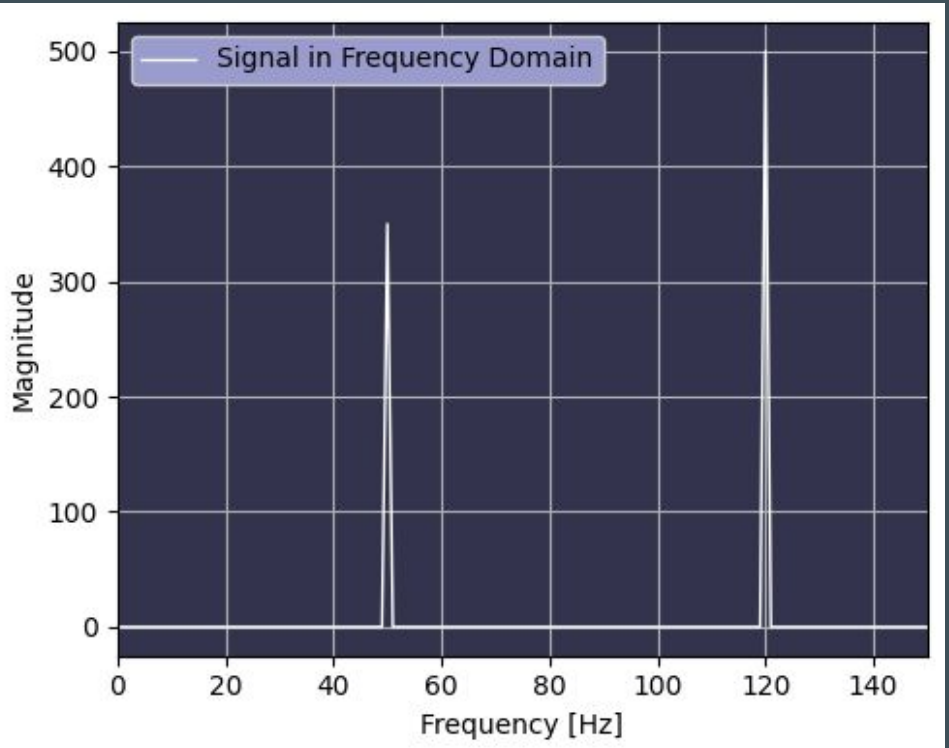
$$e^{j\theta} = \cos(\theta) + j\sin(\theta), \quad e^{-j\theta} = \cos(\theta) - j\sin(\theta)$$

Signals in Time and Frequency Domain



Time- and Frequency Domain

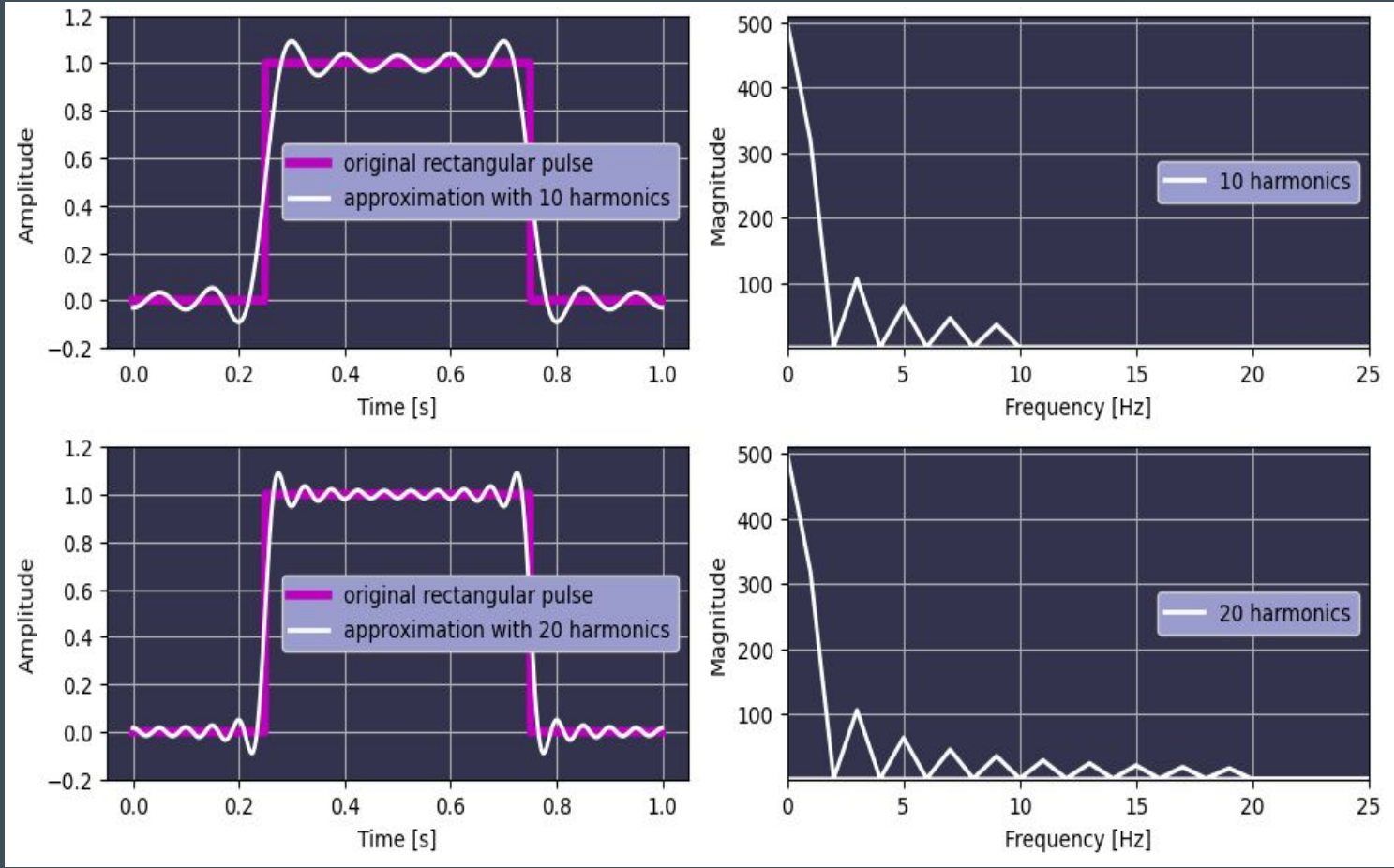




Ideal and Real-Life Rectangular Function

- In frequency domain: $\mathcal{F}\left\{\text{rect}\left(\frac{t}{T}\right)\right\} = T \cdot \text{sinc}(fT)$
  - With $\text{sinc}(x) = \frac{\sin(\pi x)}{\pi x}$
- The longer the pulse in time domain (“T”), the smaller the sinc-function in frequency domain
- A time-domain signal cannot have infinitely sharp edges, bandwidth is limited in real life

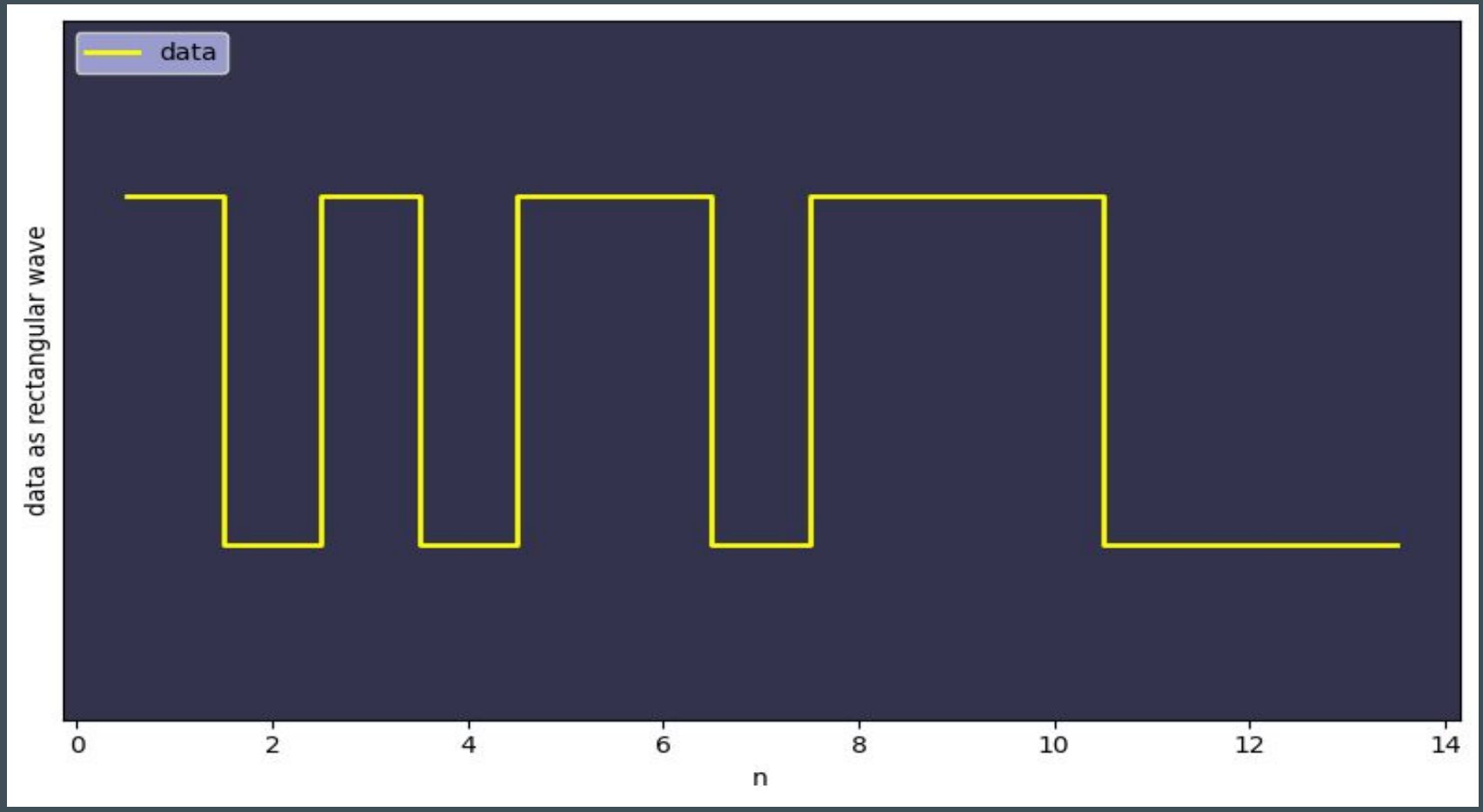
Signal Approximation in Time and Frequency



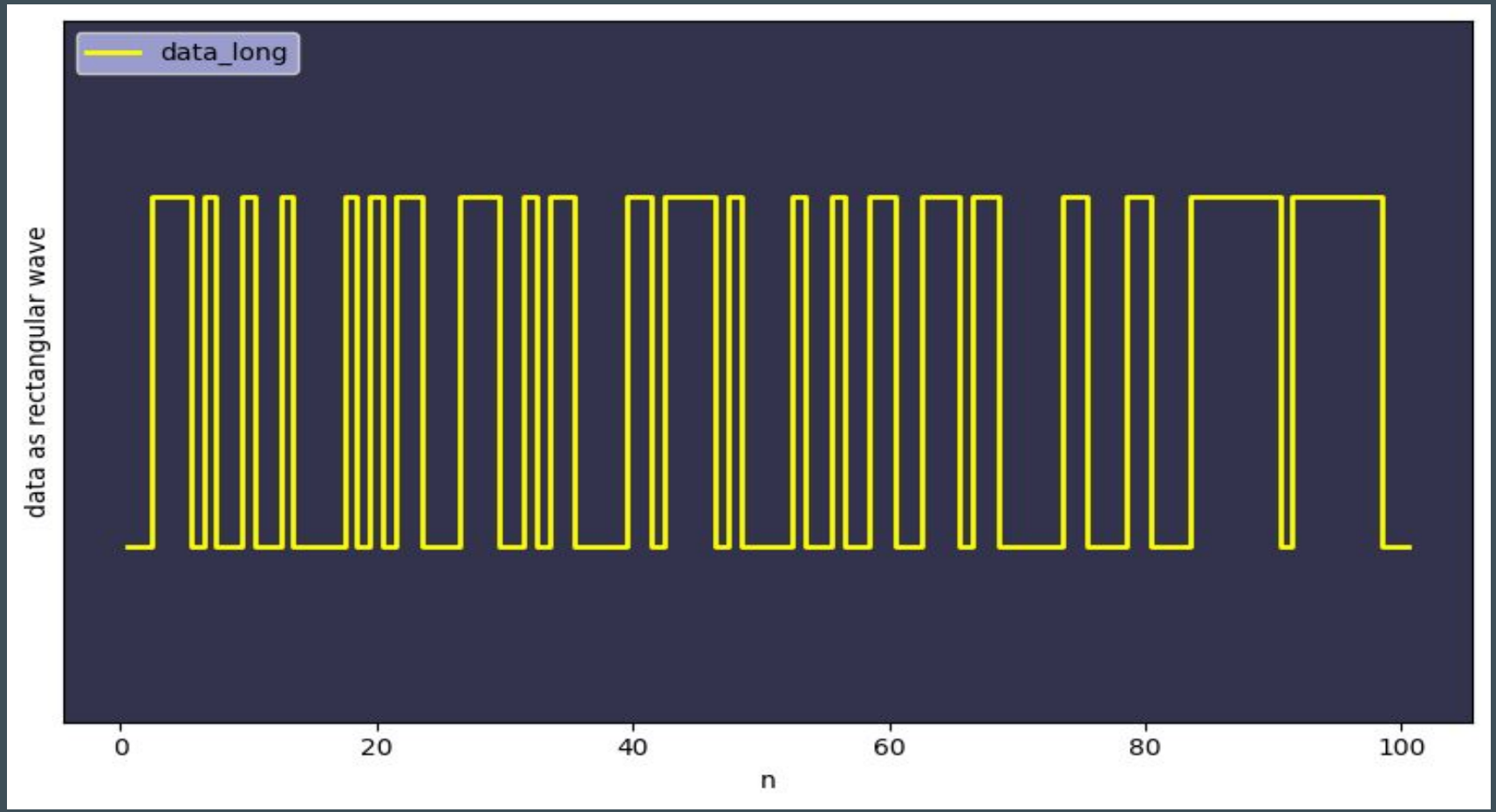
Exercise: Signal Approximation

- Open in your course folder: 02_Notebooks/01_Data_and_Signals/01.3_Fourier_Approximation_of_a_Rectangular_Pulse.ipynb
- Run the cells and answer the questions.
- Save the completed notebook and upload it to Moodle.

Data Signal



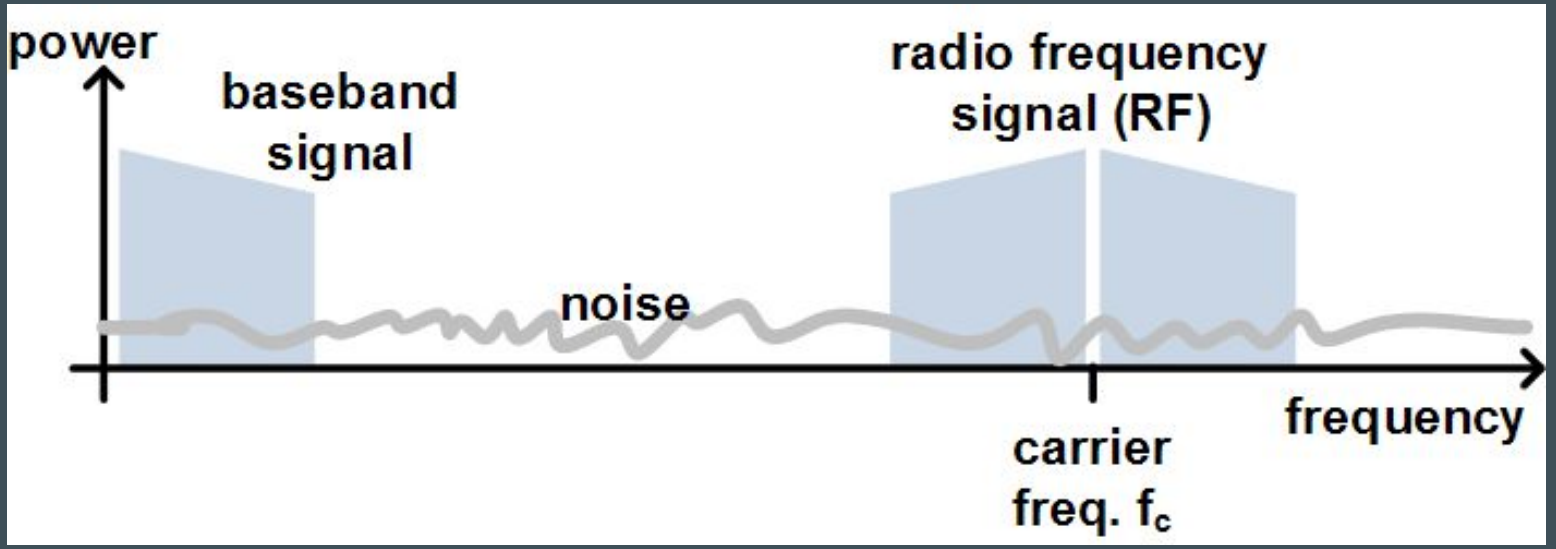
Data Signal



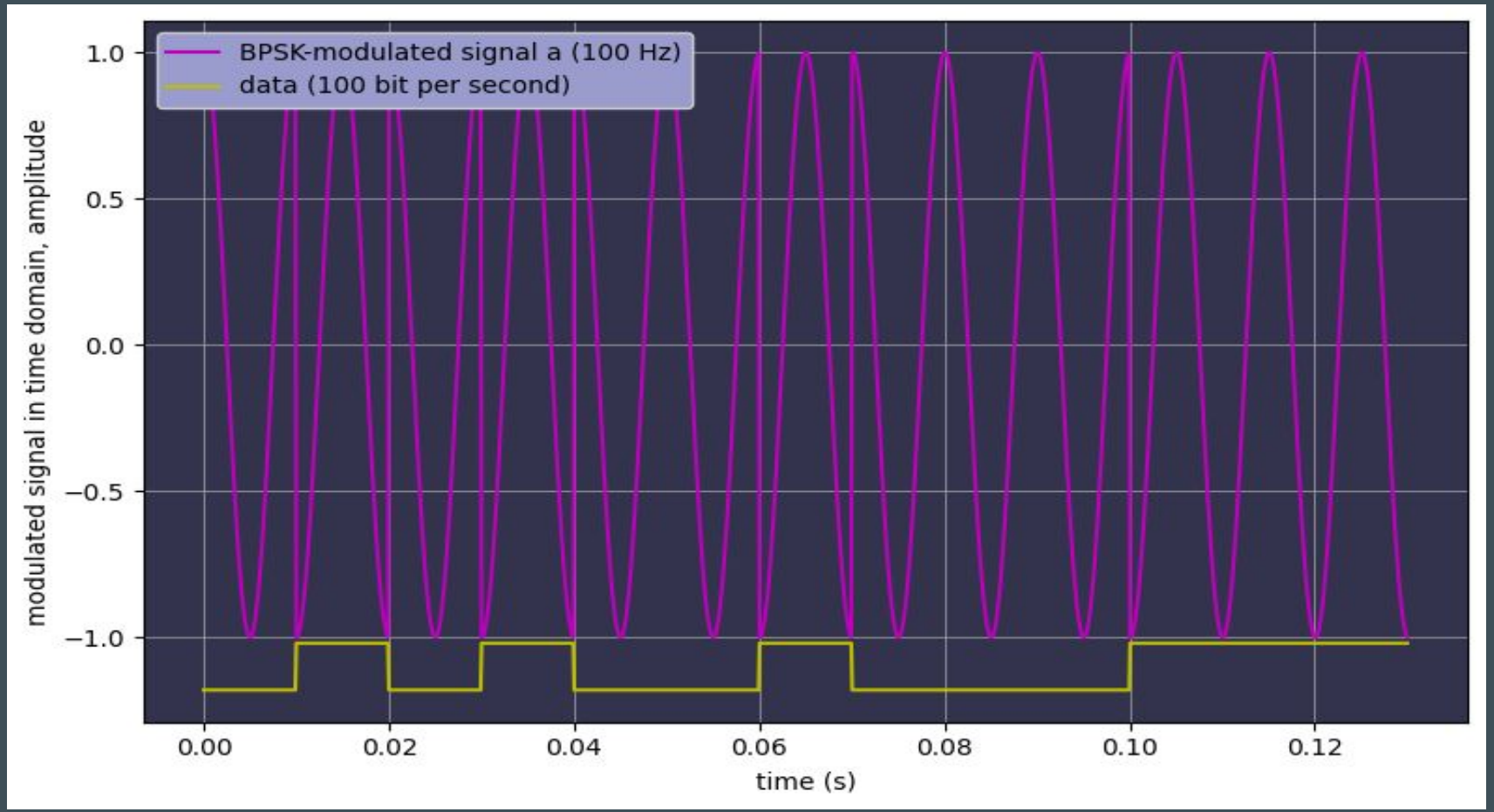
Baseband Signal, and High Frequency Carrier

- Baseband signal is modulated with incoming data
- Radio frequency signal propagates through medium

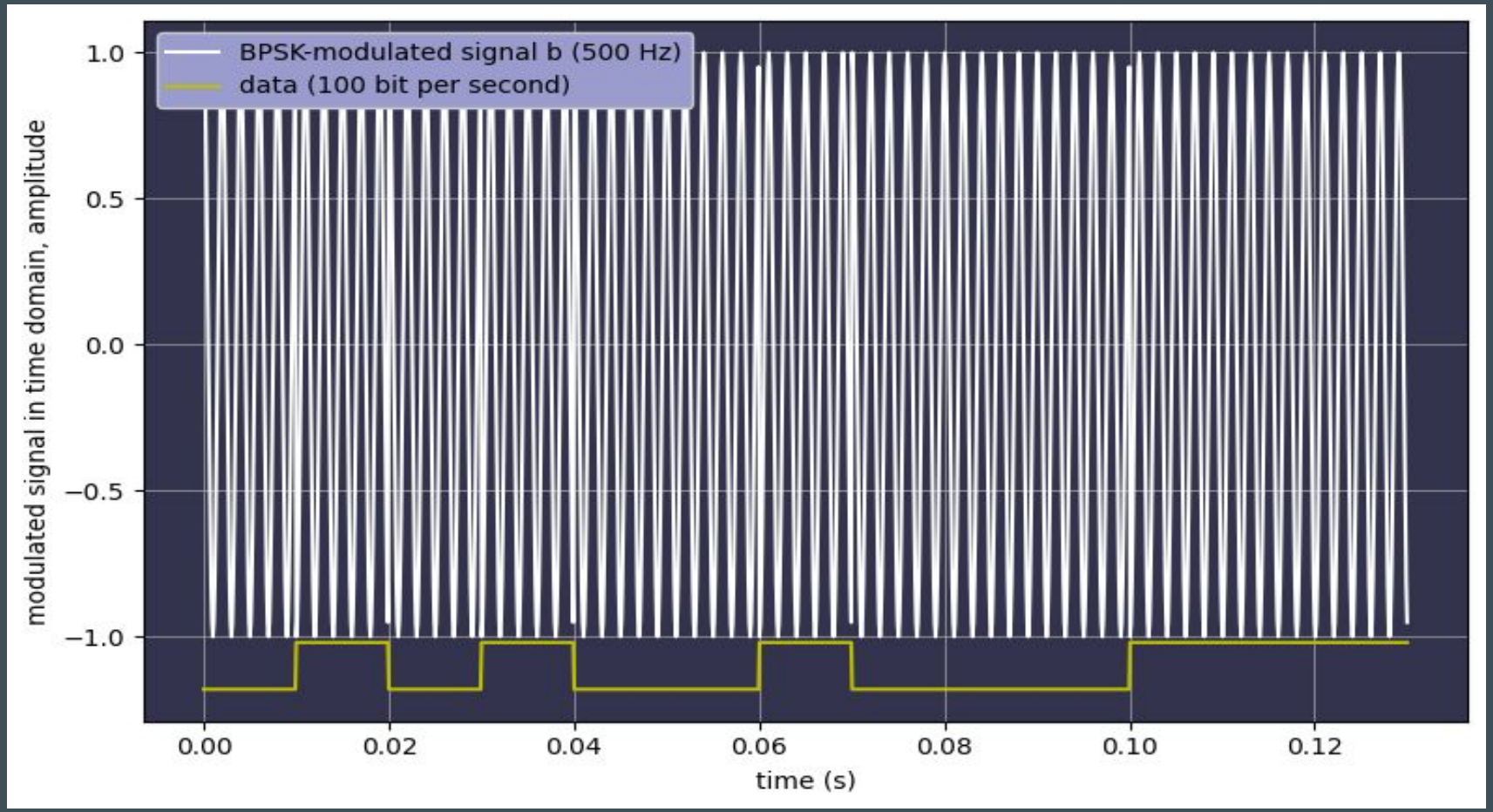
Example: 20 MHz fc = 2480 MHz



Modulation: Binary Phase Shift Keying, BPSK

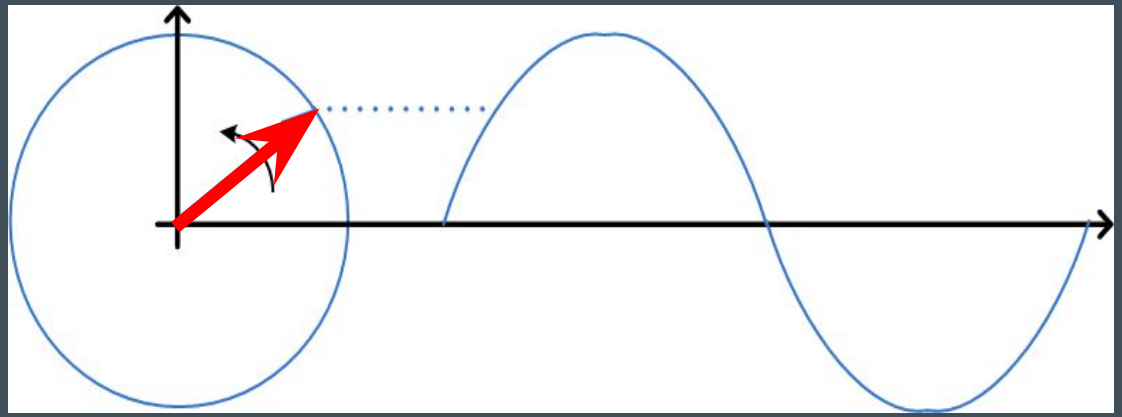


Modulation: Binary Phase Shift Keying, BPSK



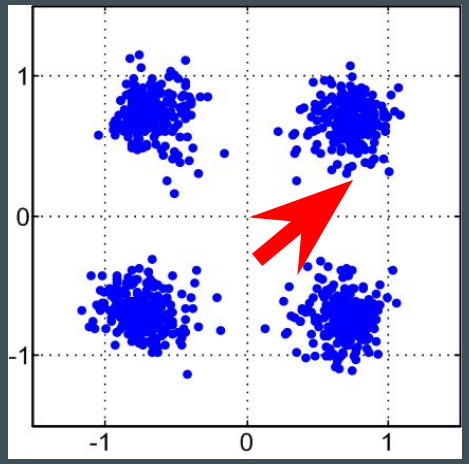
Illustrating Harmonic Waves with Vectors

- Instead of sinus waves in time, harmonic waves can also be illustrated with a vector
  - Phase & amplitude describe the baseband signal, relative to a reference
  - Speed of vector spinning around the origin, i.e., the signal frequency is ignored

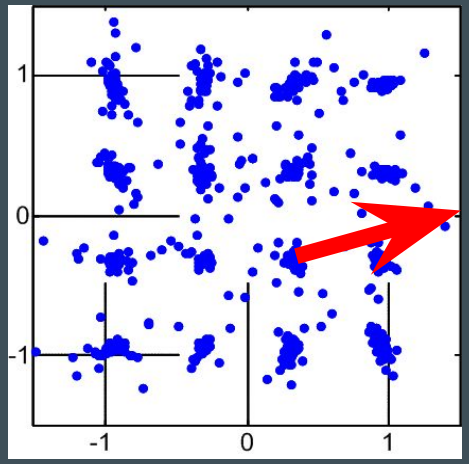


Modulation of Radio Signals

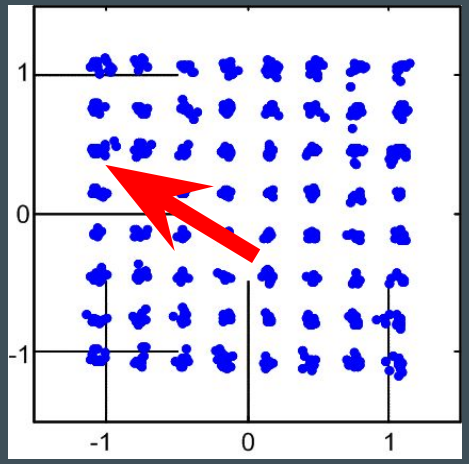
- A symbol modulated at QPSK carries 2 bit
  - 16-QAM carries 4 bit, and 64-QAM carries 6 bit
  - Higher modulation is more sensitive to noise or interference



Quadrature Phase Shift Keying (QPSK)

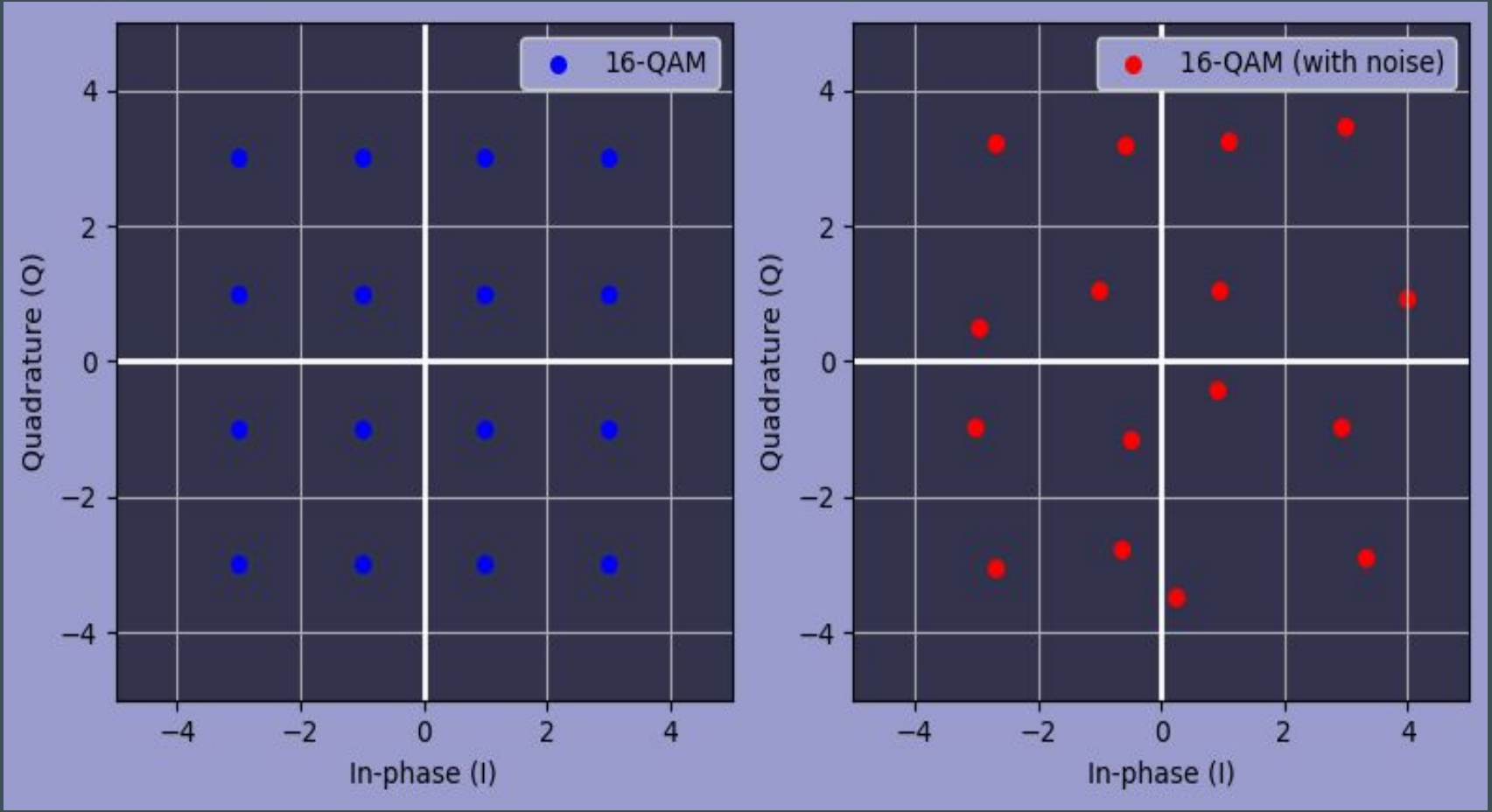


16 Quadrature Amplitude Modulation (16-QAM)

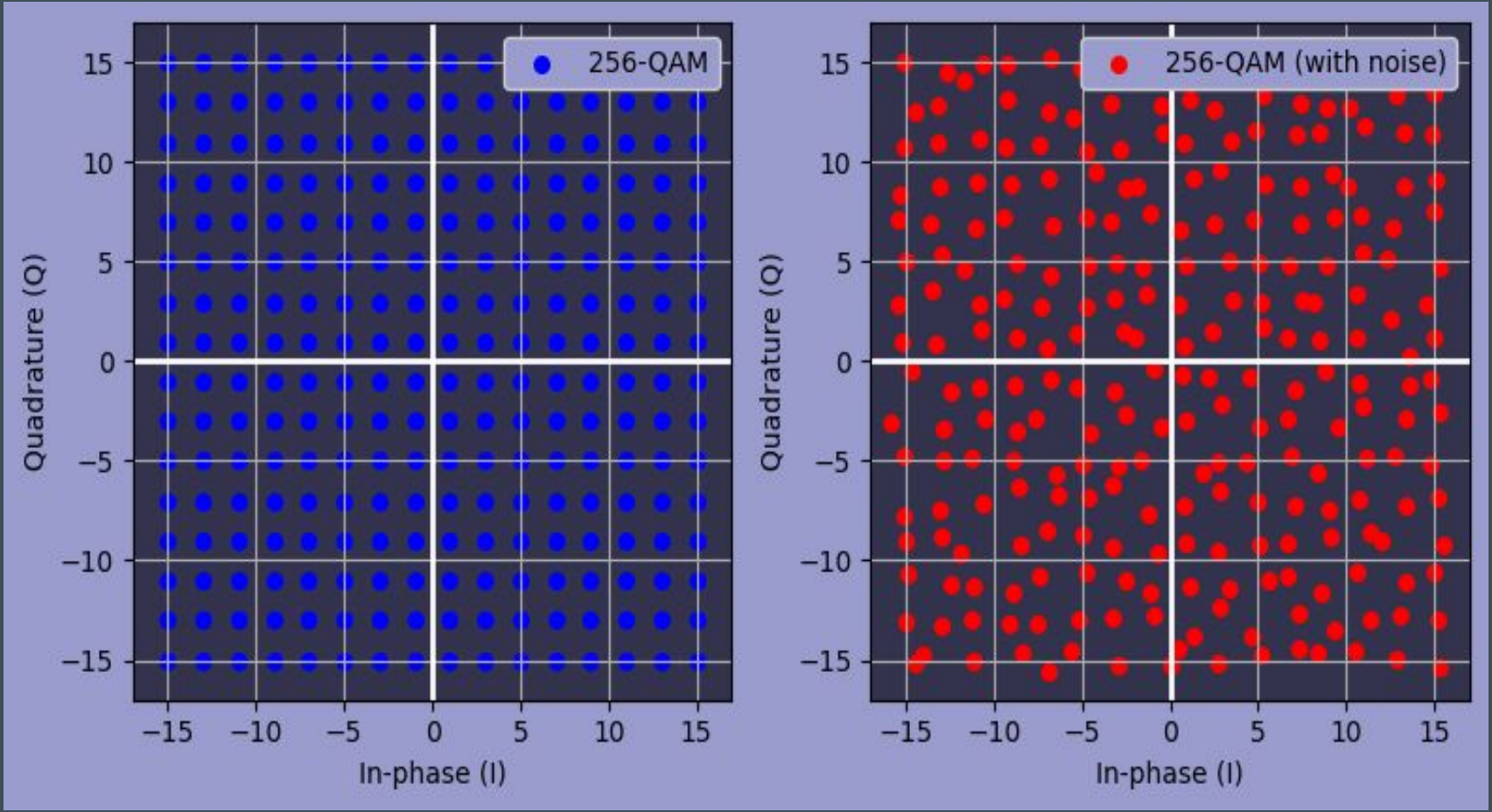


64 Quadrature Amplitude Modulation (64-QAM)

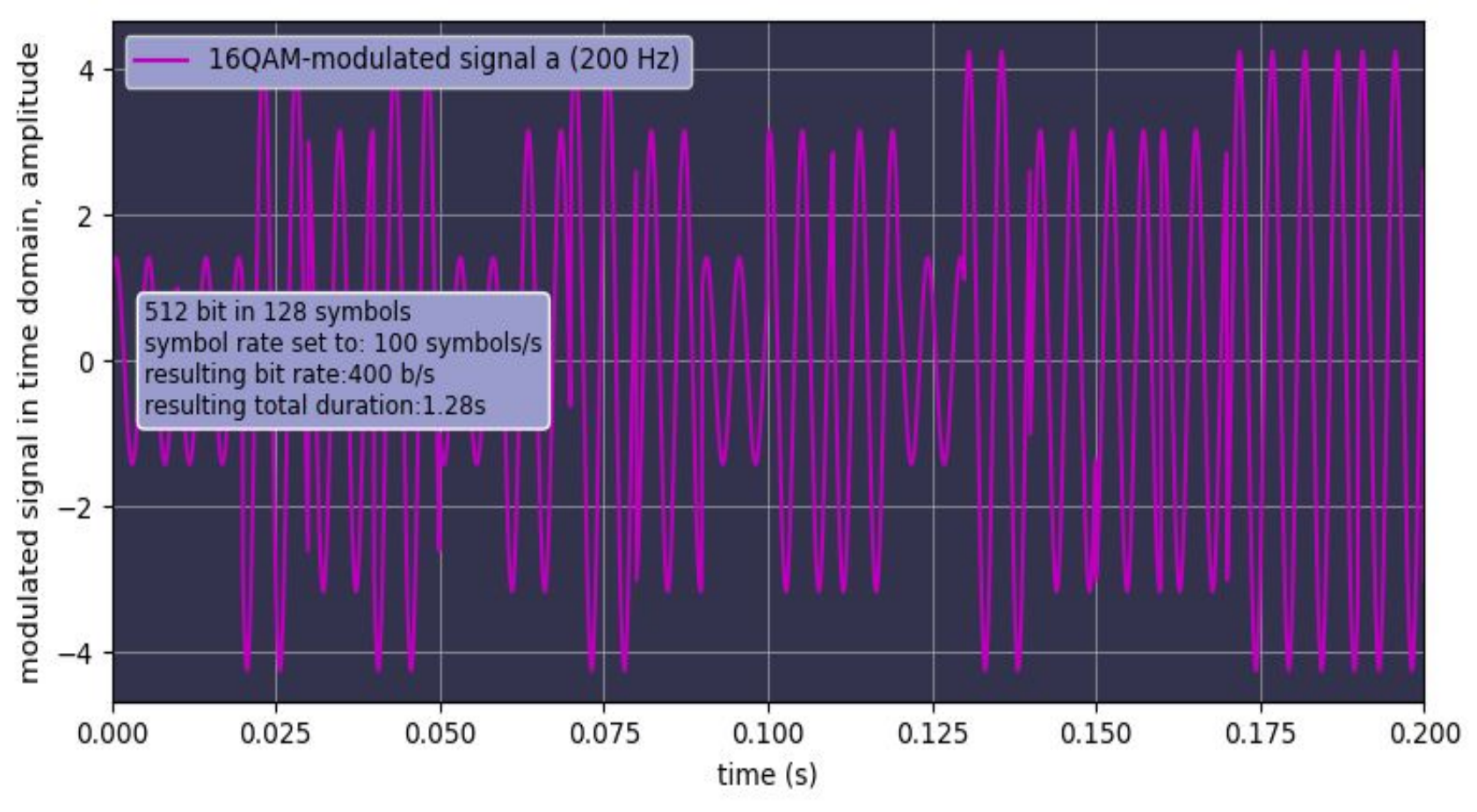
16 QAM


16-QAM
Quadrature (Q)
In-phase (I)
16-QAM (with noise)
Quadrature (Q)
In-phase (I)

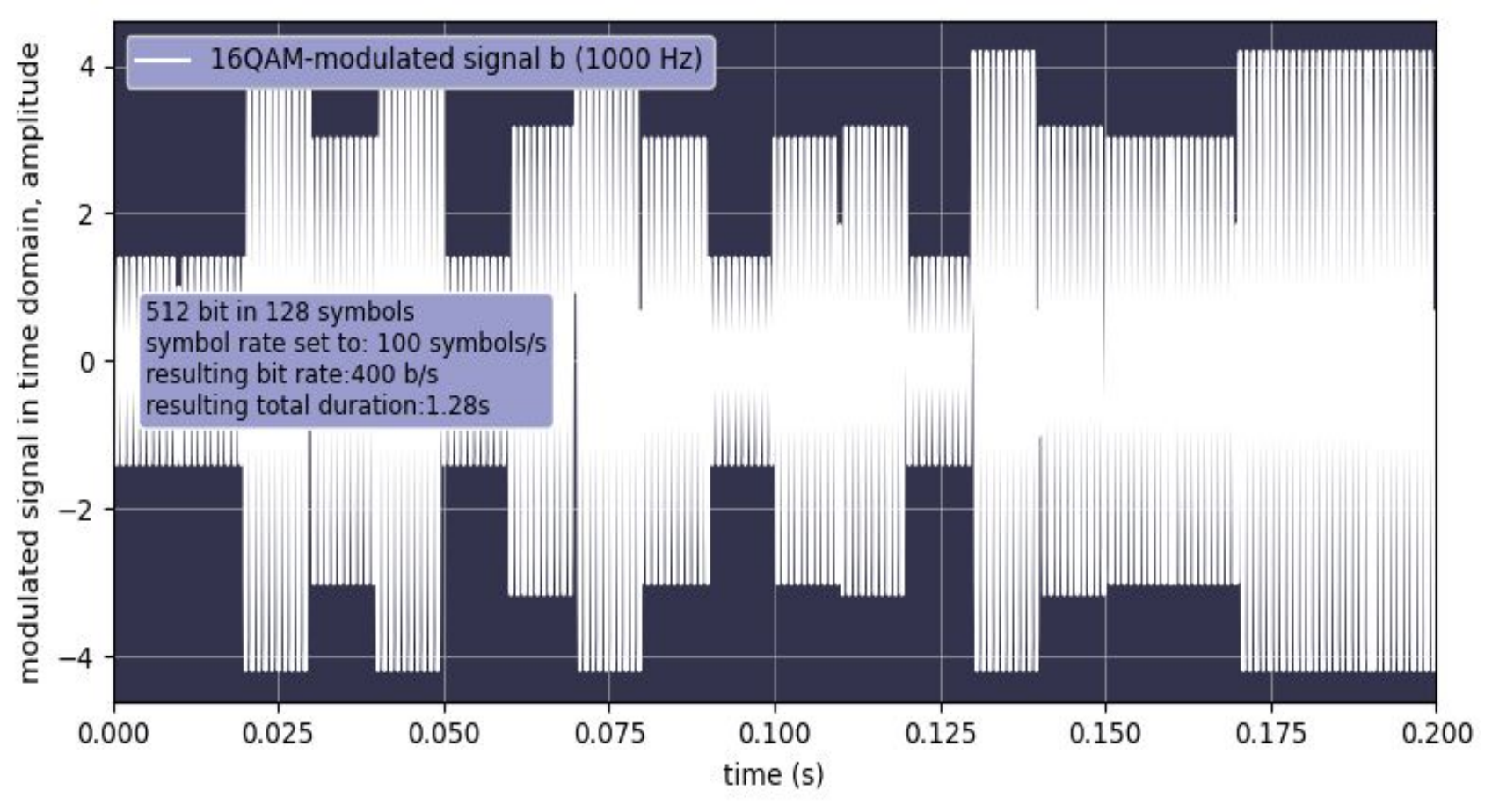
256 QAM


256-QAM
256-QAM (with noise)
Quadrature (Q)
In-phase (I)
15
10
5
0
−5
−10
−15

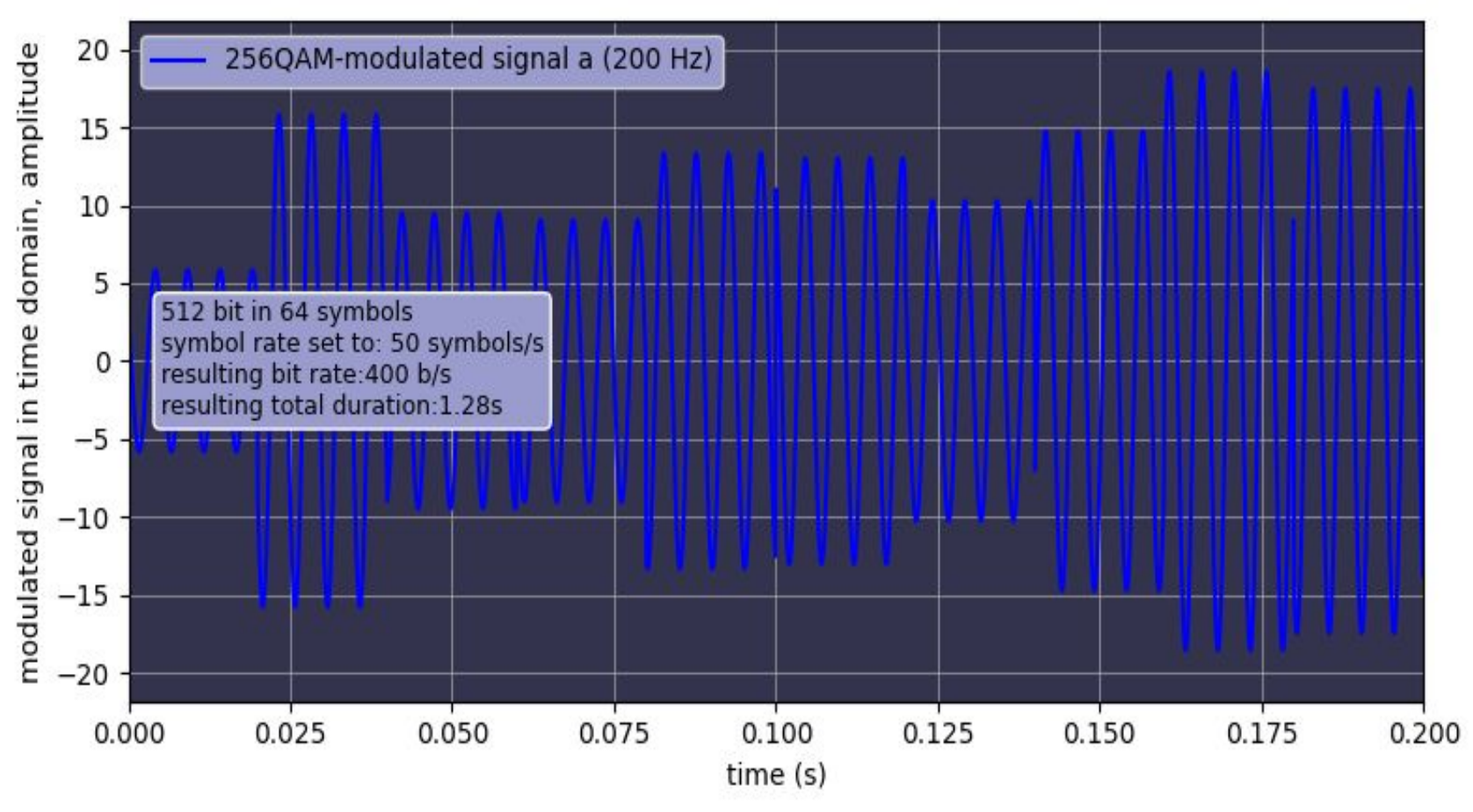
16 QAM in time



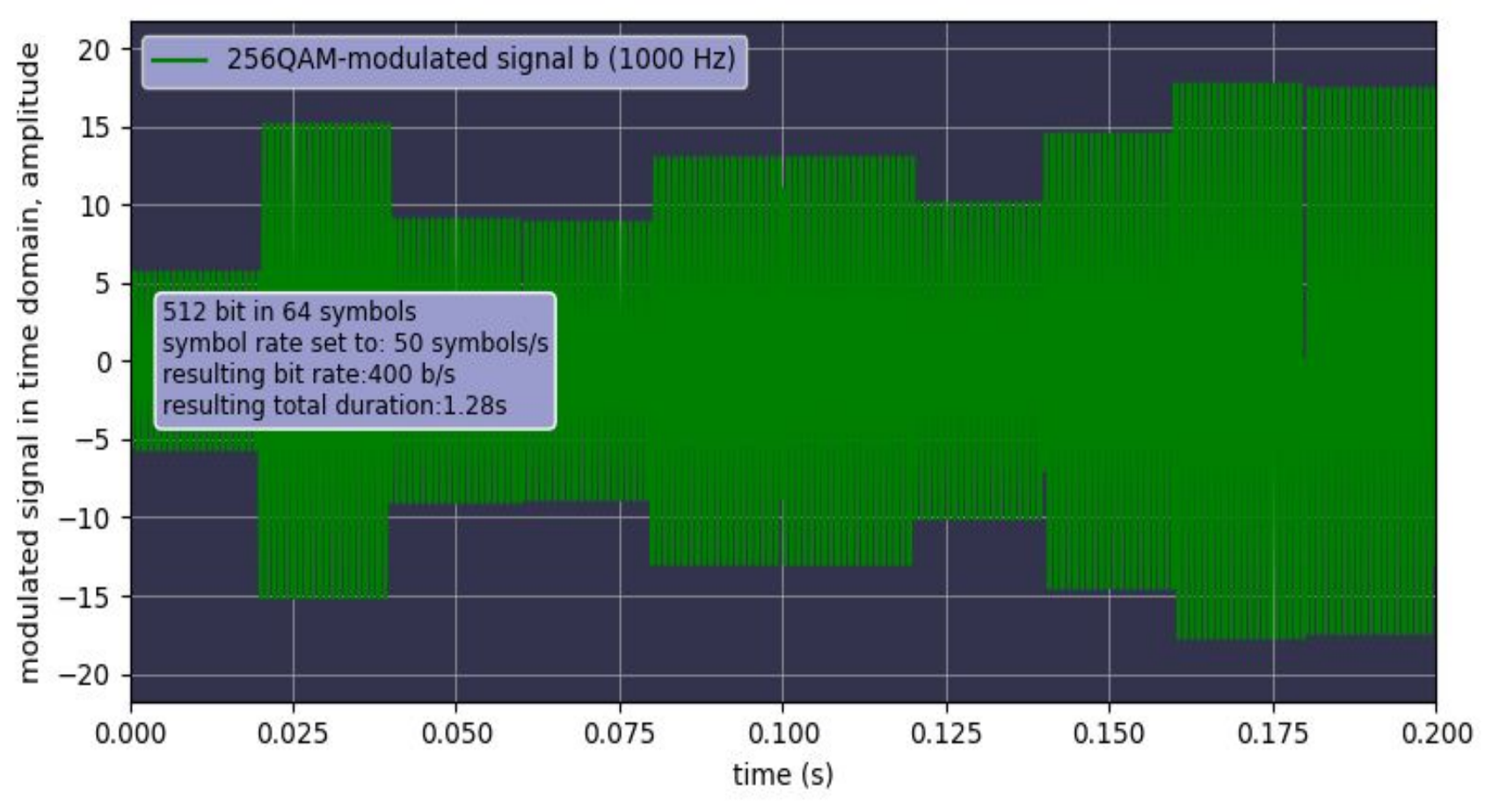
16 QAM in time



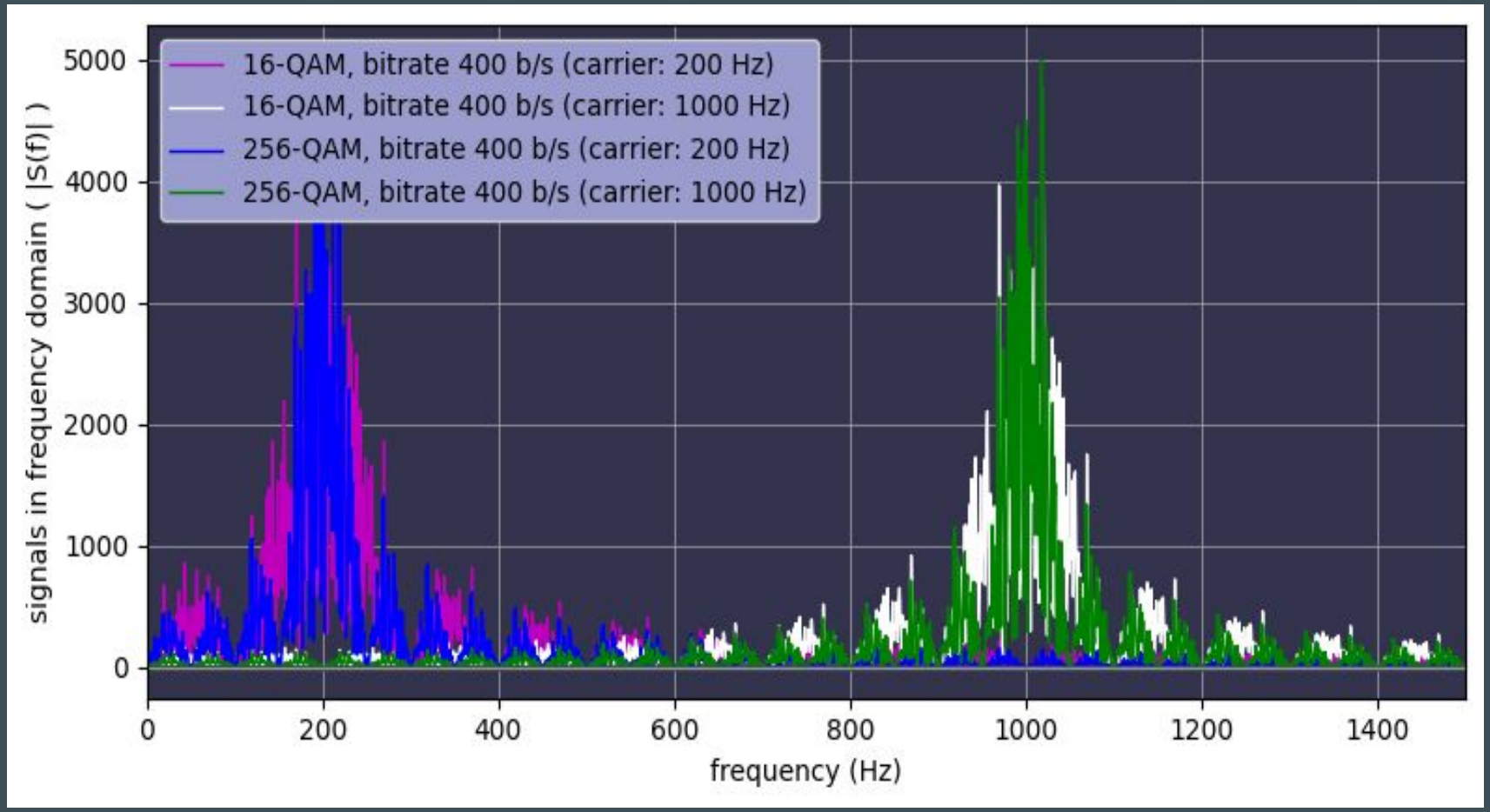
256 QAM in time



256 QAM in time



Frequency Domain



Exercise: Radio Spectrum

- Open in your course folder: 02_Notebooks/02_Data_Communication/02.8_Radio_Spectrum.ipynb
- Run the cells and answer the questions.
- Save the completed notebook and upload it to Moodle.

Noise and Maximum Channel Capacity

$$C = B \log_2 \left(1 + \frac{P_s}{P_n}\right) \qquad P_n = N = k_B T B$$

- C = channel capacity in bits per second (b/s)
- B = bandwidth in Hertz (Hz)
- Ps = signal power
- Pn = N = thermal noise power in Watt
- k_B = Boltzmann's constant (1.38×10−23 J/K)
- T = temperature in Kelvin (K)

Ps = Pn $\Rightarrow$ C = B

Spectrum Requirements

- We see that there is a benefit in using higher modulation:
  - More data can be transmitted with less channel bandwidth
- However, the maximum bit per symbol (b/s) depends on the received signal power relative to noise and interference
  - The benefit of lower modulation is the robustness against noise

Spectrum Requirements

- Signal bandwidth (in Hertz, Hz) cannot be larger than the allocated channel bandwidth
  - This limits the system sampling rate, aka the symbol duration
- Signal power cannot be higher than allowed by regulation
  - This limits the distance between transmitter and receiver

Optical vs. Radio

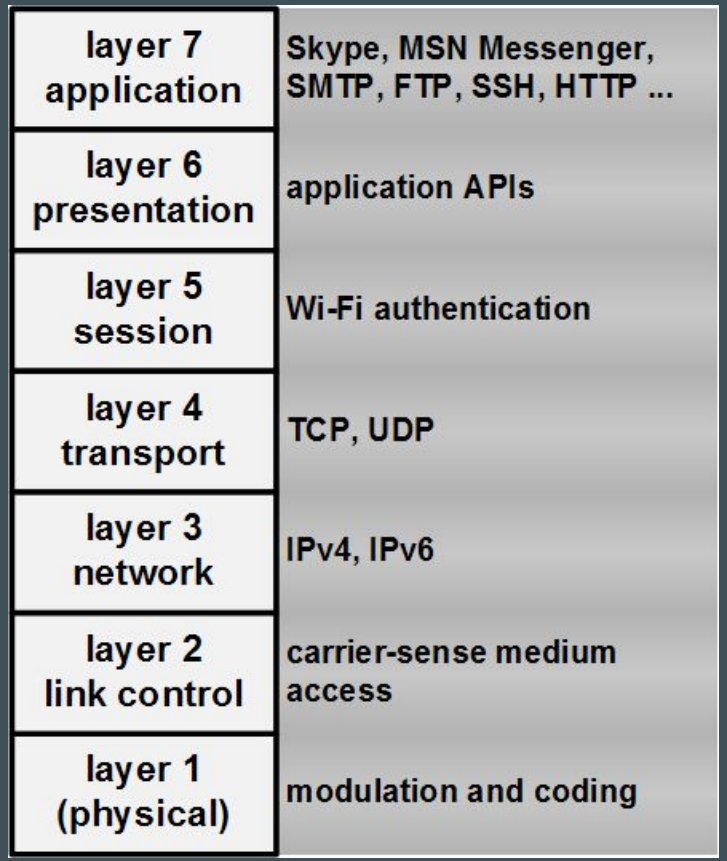
- The bandwidth of infrared and visible light spectrum is very large (1000 GHz), higher data rates are available
- Laser: coherent and monochromatic light, light waves are in phase both spatially and temporally
- LED: incoherent light, waves in a wider range of phases and directions, leading to a broader, less focused beam

Exercise: Digital Modulation

- Open in your course folder: 02_Notebooks/02_Data_Communication/02.4_Digital_Modulation.ipynb
- Run the cells and answer the questions.
- Save the completed notebook and upload it to Moodle.

ISO/OSI Reference Model - Layer 1 and Layer 2

- Everything discussed so far was related to the layer 1 of the seven Layer ISO/OSI reference model
- Layer 2 aspects are in the following discussed
  - ISO: International Standardizations Organization
  - OSI: Open Systems Interconnection



Forward Error Correction, FEC

- Source (transmitter) sends redundant information in addition to the data
  - Simple example: repeat everything that was sent
- Destination (receiver) decodes the data that does not contain any errors (some errors can be corrected)
- FEC is most relevant when the receiver does not request a retransmission (example: Broadcast, or transport protocol UDP)
  - Typically, 10% ... 50% redundancy

(continued) Forward Error Correction, FEC

- Transmitter adds redundancy, checksum
  - Predetermined scheme, known to transmitters and receivers
- Receiver uses redundancy to reconstruct data
  - Block by block (or continuously with convolutional coding)
  - Block codes work on fixed-size blocks (packets) of bits or symbols of predetermined size
  - Convolutional codes such as the well-known “Viterbi”code family work on bit or symbol streams of arbitrary length
- Modern systems such as 5G or Wi-Fi 6 use combinations of both coding schemes

Example: (8,4)-Hamming Coding

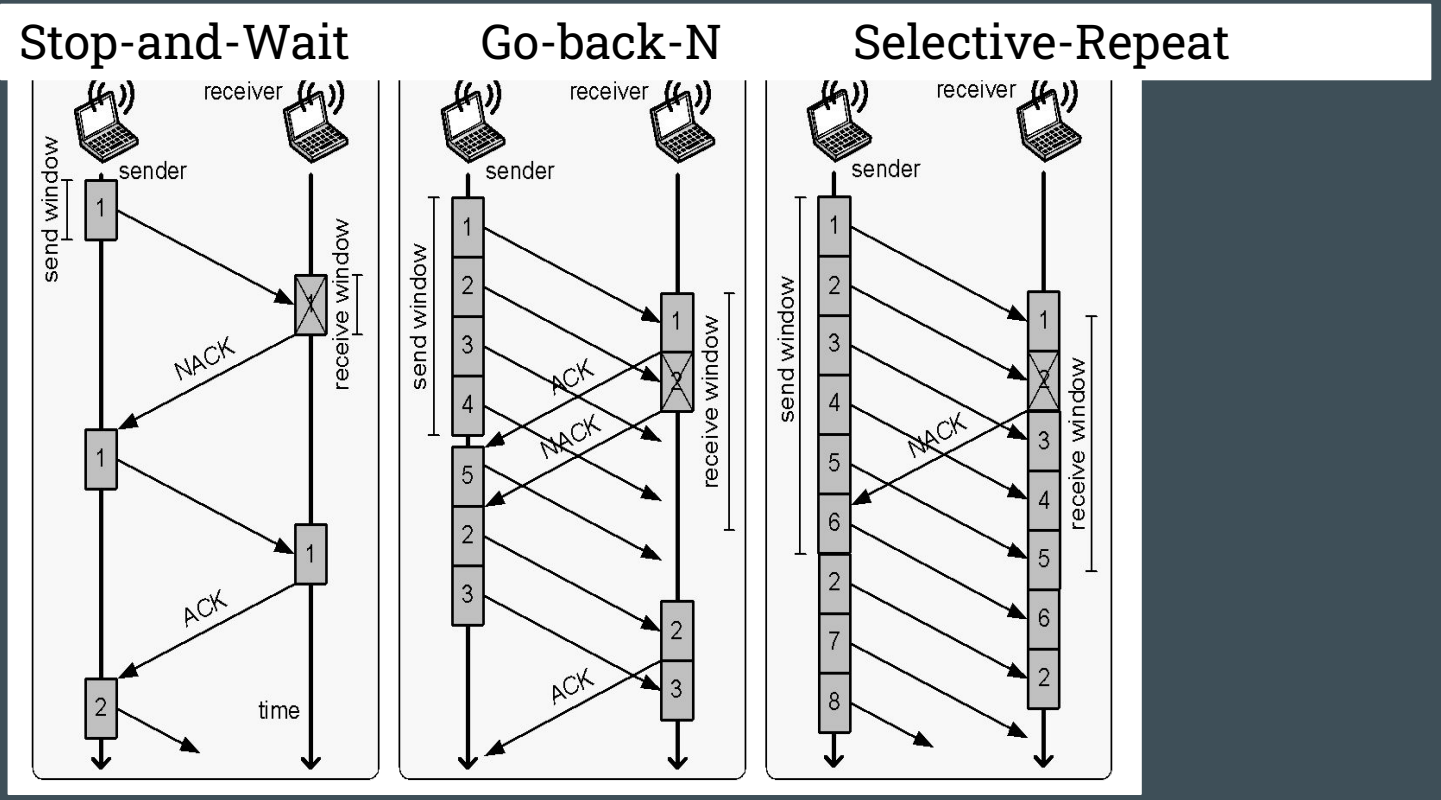
- Bit s1...s3 determine parities of the four data bit 1...4
- s4 determines the overall parity
  - Single bit errors can be detected and corrected
  - Double bit errors can be detected



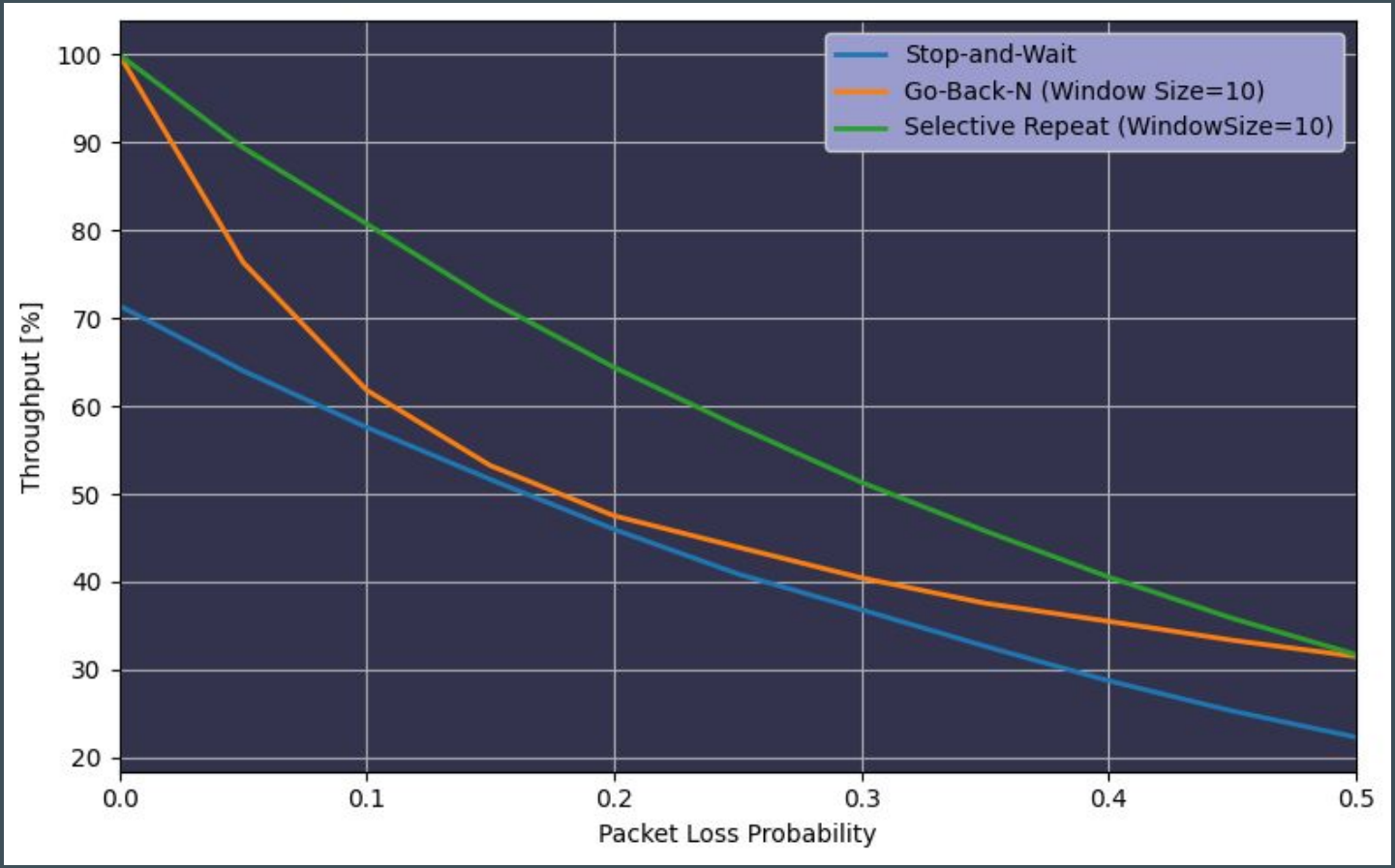
d1	d2	d3	d4	s1	s2	s3	s4	
0	0	1	1	1	0	1	0	original tx
0	0	1	1	1	1	1	0	can be corrected
0	0	1	0	1	0	1	0	can be corrected

Automatic Repeat Request, ARQ

- Also “Backward Error Detection and Correction”



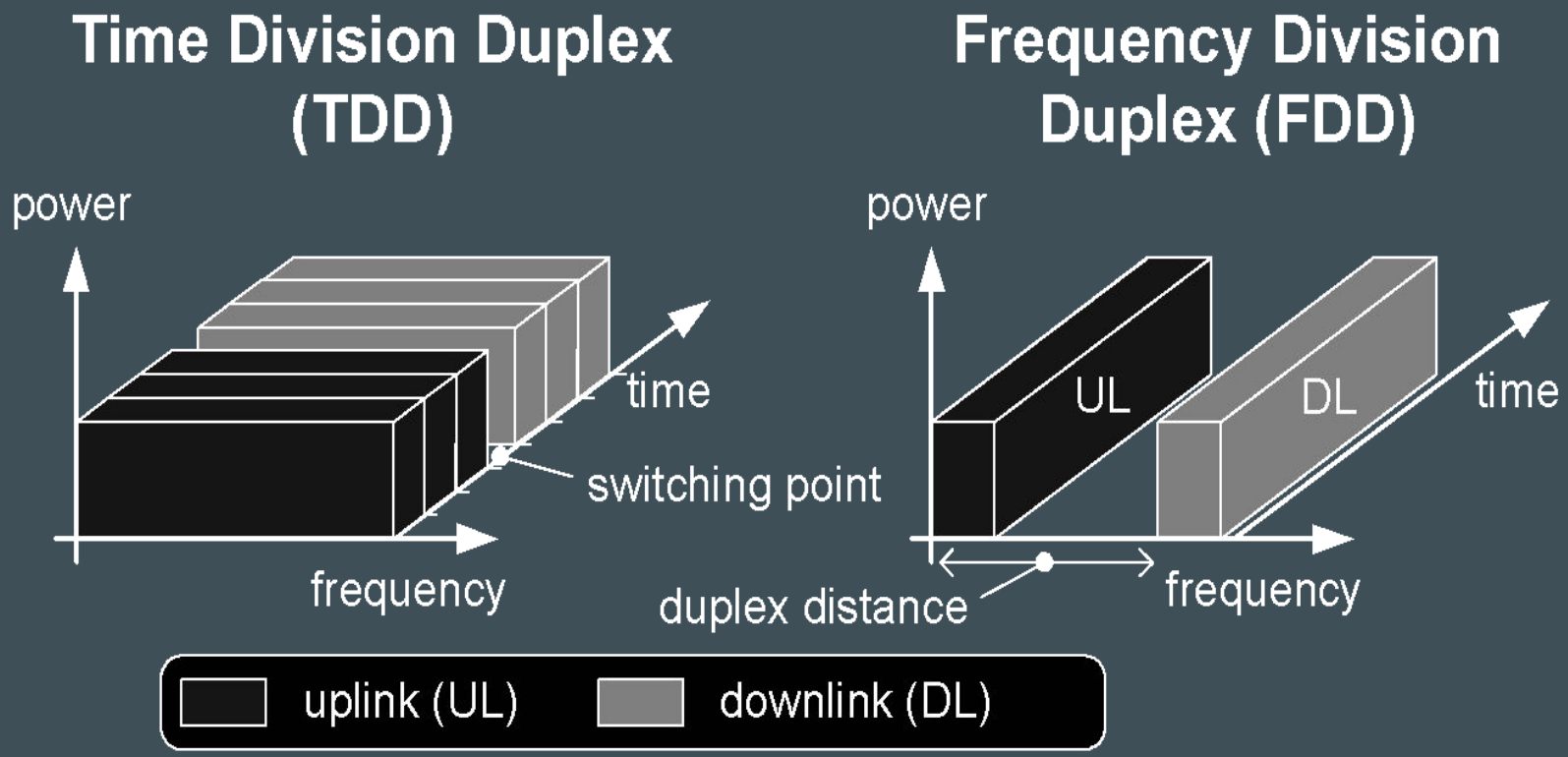
ARQ Throughput



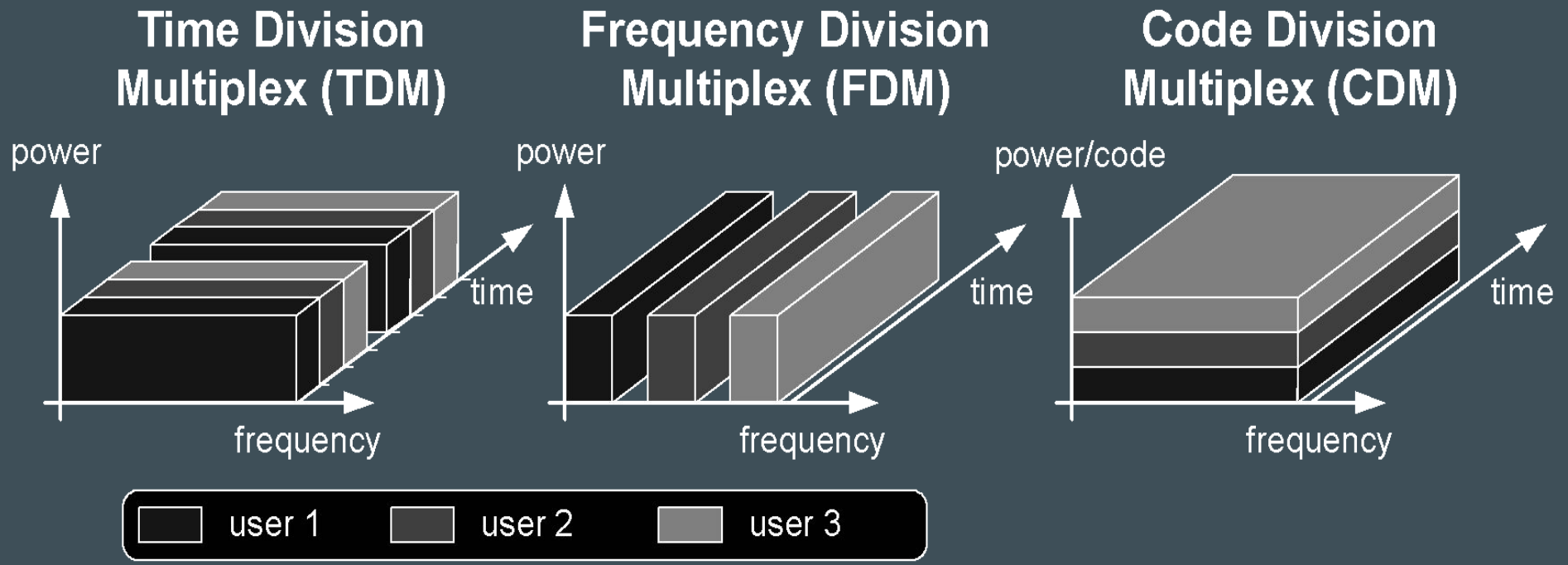
Duplexing Scheme

- Time Division Duplex
  - Switching between transmitters separates a single frequency channel, repeatedly, into uplink (UL) and downlink (DL) period
  - Ideal for asymmetric data transmission
- Frequency Division Duplex
  - DL and UL use different frequency channels of the same bandwidth
  - Ideal for symmetric services (voice), but requires paired frequency bands

Duplexing Scheme



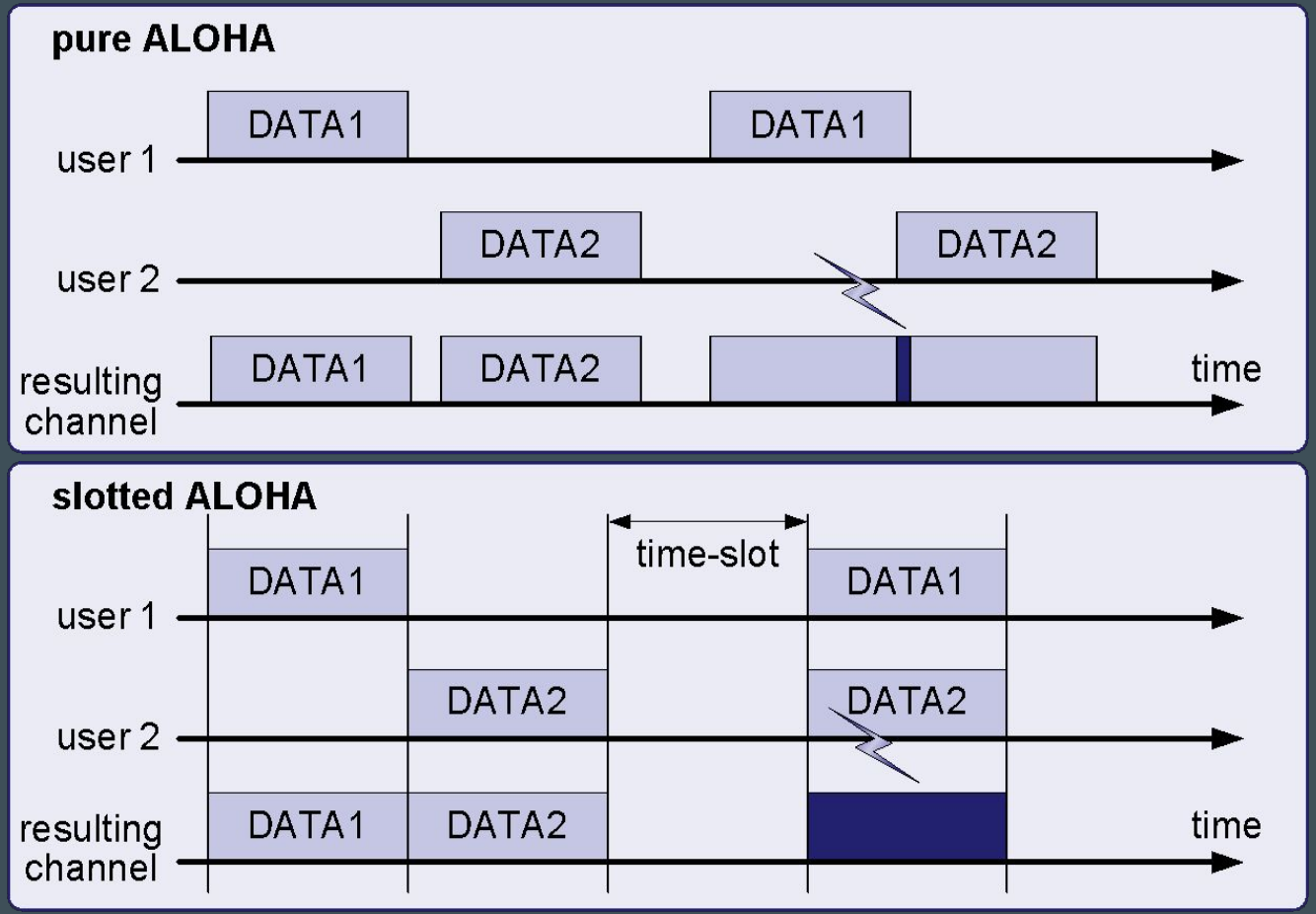
Channel Multiplexing Scheme



ALOHA and slotted ALOHA

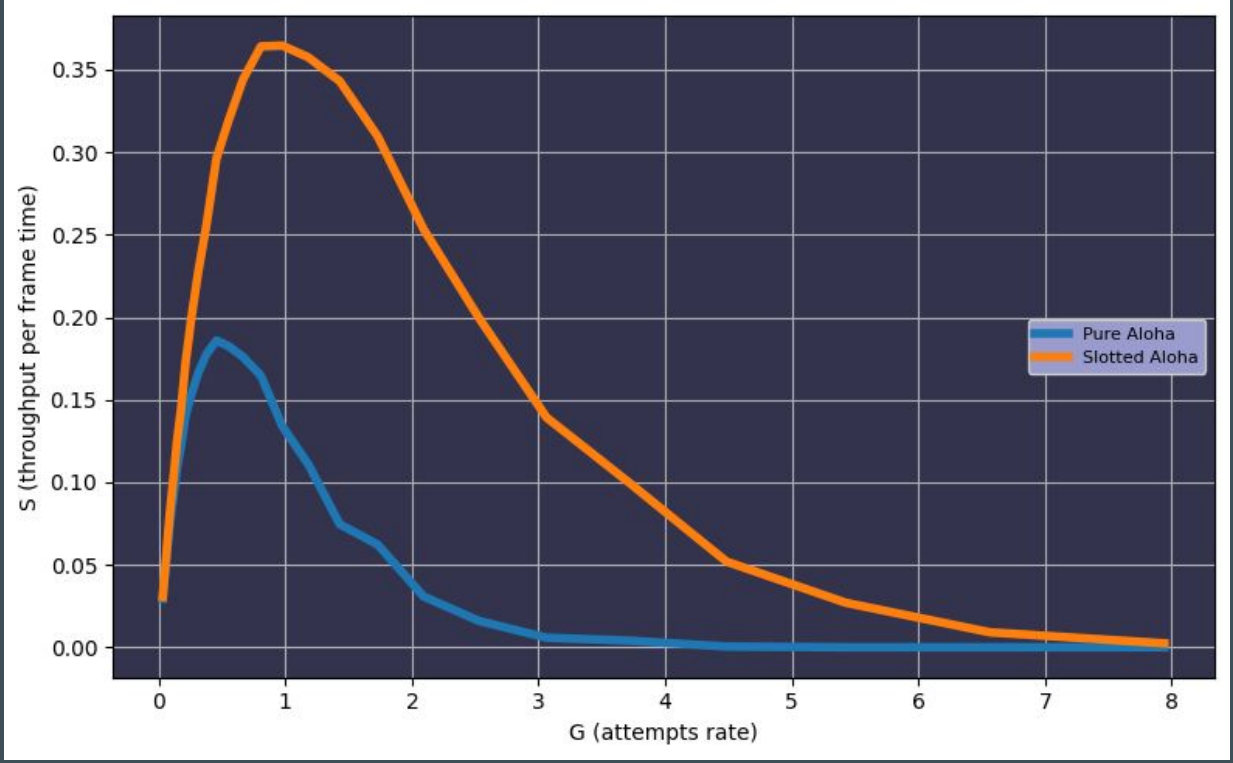
- ALOHAnet was a computer networking system developed at the University of Hawaii, 1971
- First public demonstration of a wireless packed data network, protocol “ALOHA”
  - If you have data to send, send it
  - If, while transmitting data, any data from another station is transmitted, there has been a collision. All transmitting stations will need to try resending later.
- Improvement "Slotted ALOHA," introduced slots

ALOHA and slotted ALOHA



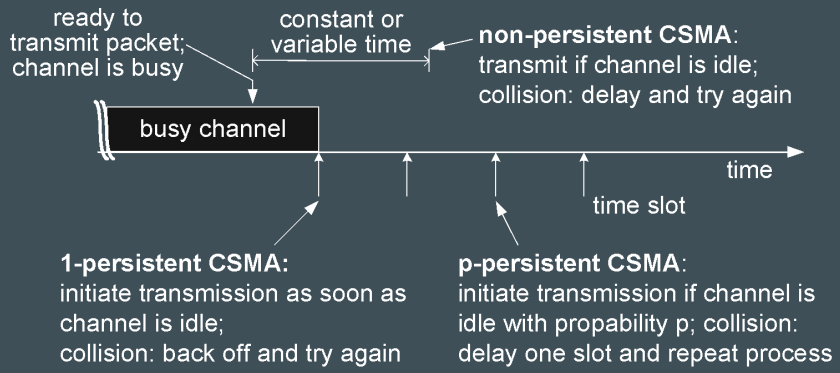
ALOHA and Slotted ALOHA Throughput

- Max throughput of slotted ALOHA is 2 x ALOHA

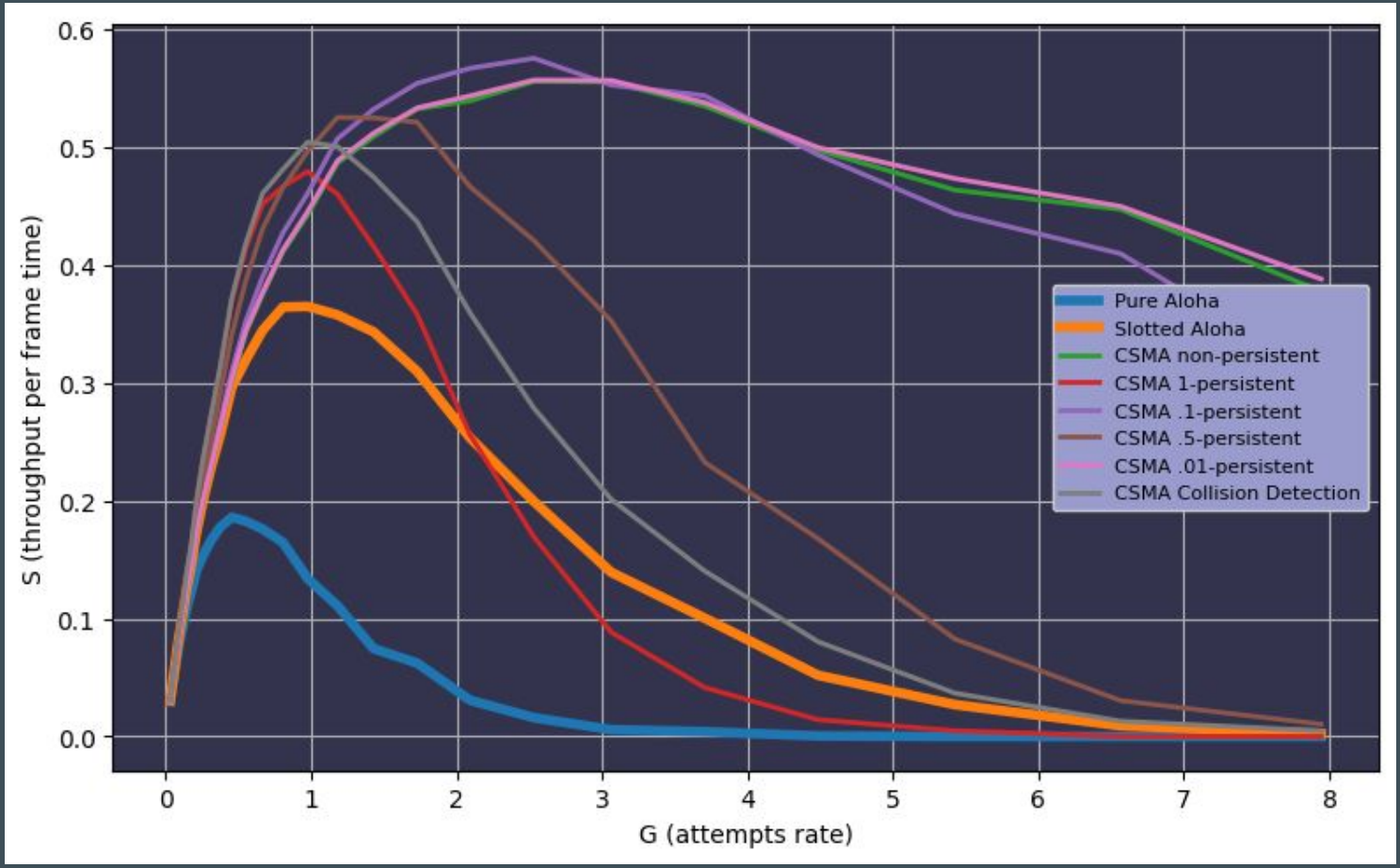


Carrier Sense Multiple Access (CSMA)

- CSMA with Collision Detection (CSMA/CD) - Ethernet
  - Jamming signal sent after detection, transmission terminated
- CSMA with Collision Avoidance (CSMA/CA) - Wi-Fi
  - stations defer from medium access when notified about transmissions



CSMA Performance (Notebook 03.5, §5.2.6)



Exercise: Wi-Fi Contention Protocol

- Open in your course folder: 02_Notebooks/03_Mobile_Systems/ 03.5_Wi-Fi_Contention_Protocol.ipynb
- Run the cells and answer the questions.
- Save the completed notebook and upload it to Moodle.

Exercise: Basics of Data Communication

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.1_Basics_of_Data_Communication.ipynb
- Trace a familiar wireless system from source to sink.
- Answer the questions on design goals and coding redundancy.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

Exercise: Source Coding

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.2_Source_Coding.ipynb
- Compare text entropy, Huffman coding, and JPEG compression.
- Complete Section 2.7 and explain the compression trade-offs.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

Exercise: Channel Coding

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.3_Channel_Coding.ipynb
- Explore repetition coding and majority decoding.
- Complete Section 3.5; compare reliability with code rate.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

Exercise: Noise and Interference

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.6_Noise_and_Interference.ipynb
- Explore how noise and SNR affect bit errors.
- Complete Section 6.6, including the thermal-noise calculation.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

Exercise: Path Loss of Radio Communication

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.7_Path_Loss_of_Radio_Communication.ipynb
- Explore path loss, receiver sensitivity, and the link budget.
- Complete Section 7.5 using the drone-link example.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

Exercise: Shannon Capacity

- Open in your course folder:
  02_Notebooks/02_Data_Communication/
  02.9_Shannon_Capacity.ipynb
- Compare noisy constellations and calculate channel capacity.
- Complete Section 9.4 on SNR, bandwidth, and reliable rate.
- Save your responses and submit via Moodle
  as instructed in the assignment guidelines.

